

MAT-130 : Integral Calculus

Assignment:0

A list of formulas for differentiation and integration from Chapters 2, 3 and 5.

Section 7.1: Overview

MAT 130: 7.1 → MAT 120: 5.3 : Integration by Substitution

Method: Integration by Substitution [u-substitution]

Recall Chain Rule: $\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$; $[\frac{d}{dx}(f(u)) = f'(u) \cdot \frac{du}{dx}]$

For example, $\frac{d}{dx}[\sin^2(e^{\sqrt{2x}})] = \frac{d}{dx}[\sin(e^{\sqrt{2x}})]^2 = 2 \sin(e^{\sqrt{2x}}) \cdot \cos(e^{\sqrt{2x}}) \cdot e^{\sqrt{2x}} \cdot \frac{1}{2\sqrt{2x}} \cdot 2$

Backward Chain Rule: $\int f'(g(x)) \cdot g'(x) dx$.

The method will work if there is

- ➔ A product of 2-functions
- ➔ 1-composite function
- ➔ Other one is the derivative of the input of the composite function up to a scalar multiplication.

Understanding: Identify the composite function. Set the input function as u , **whose derivatives is given** (up to a scalar multiple is given).

Now, $\int f'(g(x)) \cdot g'(x) dx$; Set $u = g(x)$. Then $\frac{du}{dx} = g'(x)$, than is, $du = g'(x)dx$.

Then, $\int f'(g(x)) \cdot g'(x) dx$

$= \int f'(u) du = f(u) + C = f(g(x)) + C$, where C is a constant.

Examples

$$(1) \int \frac{1}{\sqrt{x}} \sec^2(\sqrt{x}) dx$$

$$= 2 \int \frac{1}{2\sqrt{x}} \sec^2(\sqrt{x}) dx ;$$

$$\text{Set } u = \sqrt{x} \text{ implies } \frac{du}{dx} = \frac{1}{2\sqrt{x}} . \text{ That is, } du = \frac{1}{2\sqrt{x}} dx$$

Now,

$$\begin{aligned} \int \frac{1}{\sqrt{x}} \sec^2(\sqrt{x}) dx &= 2 \int \frac{1}{2\sqrt{x}} \sec^2(\sqrt{x}) dx = 2 \int \sec^2(\sqrt{x}) \frac{1}{2\sqrt{x}} dx \\ &= 2 \int \sec^2(u) du = 2 \tan u + C = 2 \tan(\sqrt{x}) + C ; [\text{Formula : } \int \sec^2 x dx = \tan x + C] \end{aligned}$$

$$(2) \int \frac{x}{\sqrt{1-x^4}} dx = \int \frac{x}{\sqrt{1-(x^2)^2}} dx$$

$$\text{Set } u = x^2, \quad du = 2x dx . \text{ So, } x dx = \frac{1}{2} du$$

$$\text{Now, } \int \frac{x}{\sqrt{1-x^4}} dx = \int \frac{x}{\sqrt{1-(x^2)^2}} dx$$

$$= \int \frac{1}{\sqrt{1-(x^2)^2}} x dx$$

$$= \int \frac{1}{\sqrt{1-u^2}} \frac{1}{2} du$$

$$= \frac{1}{2} \int \frac{1}{\sqrt{1-u^2}} du$$

$$= \frac{1}{2} \sin^{-1} u + C$$

$$= \frac{1}{2} \sin^{-1}(x^2) + C$$

$$(3) \int \frac{1}{x(\ln x)^3} dx$$

$$\text{Set } u = \ln x, \quad \text{then } \frac{du}{dx} = \frac{1}{x}, \quad \text{that is, } du = \frac{1}{x} dx$$

$$\int \frac{1}{x(\ln x)^3} dx = \int \frac{1}{(\ln x)^3} \cdot \frac{1}{x} dx$$

$$= \int \frac{1}{u^3} du = \int u^{-3} du = \frac{u^{-3+1}}{-3+1} + C$$

$$= -\frac{1}{2} u^{-2} + C = -\frac{1}{2u^2} + C$$

$$= -\frac{1}{2(\ln x)^2} + C$$

$$(4) \quad \int \frac{x^2}{\sqrt{1-x^6}} dx = \int \frac{x^2}{\sqrt{1-(x^3)^2}} dx = \int \frac{1}{\sqrt{1-(x^3)^2}} \cdot x^2 dx$$

$$\text{Set } u = x^3 \quad \text{Then } x^2 dx = \frac{1}{3} du$$

$$\int \frac{x^2}{\sqrt{1-x^6}} dx = \int \frac{1}{\sqrt{1-(x^3)^2}} \cdot x^2 dx$$

$$= \int \frac{1}{\sqrt{1-u^2}} \cdot \frac{1}{3} du$$

$$= \frac{1}{3} \sin^{-1} u + C = \frac{1}{3} \sin^{-1}(x^3) + C$$

$$\text{Formula: } \int \frac{1}{\sqrt{1-u^2}} du = \sin^{-1} u + C$$

Section 7.2: Integration by Parts

Recall Last Lecture: 7.1 Integration by Substitution

Example: Integration by substitutions works

$$\int x^2 (2 + 4x^3)^{12} dx = \frac{1}{12} \int 12 x^2 (2 + 4x^3)^{12} dx,$$

But does not work for

$$\int x (2 + 4x^3)^{12} dx.$$

Section 7.2: Integration by Parts [Backward Product Rule]

Works to integrate a product of two functions where one is a differential function. That is, you know the integration of one function.

For example,

$$\int f(x) \cdot g'(x) dx.$$

Recall Product Rule: $\frac{d}{dx} [f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$

We get, $f(x)g'(x) = \frac{d}{dx} [f(x)g(x)] - f'(x)g(x)$

$$\int f(x) \cdot g'(x) dx = \int \left[\frac{d}{dx} [f(x)g(x)] - f'(x)g(x) \right] dx$$

$$= \int \frac{d}{dx} [f(x)g(x)] dx - \int f'(x)g(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$

$$\int f(x) \cdot g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$

If we set $u = f(x)$, and $dv = g'(x)dx$, then

$$\frac{du}{dx} = f'(x) \Rightarrow du = f'(x)dx, \text{ and } \int 1 dv = \int g'(x)dx \Rightarrow v = g(x) + 0;$$

[Note: C must be 0 for this method]

Then from

$$\int f(x) \cdot g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx \text{ we get } \int u dv = uv - \int v du$$

Formulas:

Indefinite Integral: $\int u dv = uv - \int v du \dots \dots (1)$

Definite Integral: $\int_a^b u dv = [uv]_a^b - \int_a^b v du \dots \dots (2)$

LIATE

L $\rightarrow$ Logarithmic Function

I $\rightarrow$ Inverse Trigonometric Function

A $\rightarrow$ Algebraic Function, means radicals, polynomials, etc.

T $\rightarrow$ Trigonometric Function

E $\rightarrow$ Exponential Function

This order helps you to choose u and dv . The function which comes first in this order is set up as u .

Exercises:

1) $\int x^2 \ln x dx$

Set $u = \ln x$, $dv = x^2 dx$

$$\int x^2 \ln x dx = \int u dv = uv - \int v du$$

Now, $u = \ln x \Rightarrow \frac{du}{dx} = \frac{1}{x}$, that is, $du = \frac{1}{x} dx$

and $dv = x^2 dx \Rightarrow \int 1 dv = \int x^2 dx$, that is, $v = \frac{x^3}{3}$; [For this method, the constant $C = 0$]

$$\begin{aligned} \text{So, } \int x^2 \ln x dx &= \int u dv = uv - \int v du \\ &= \ln x \cdot \frac{x^3}{3} - \int \frac{x^3}{3} \frac{1}{x} dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^2 dx \\
&= \frac{1}{3} x^3 \ln x - \frac{1}{3} \cdot \frac{x^3}{3} + C
\end{aligned}$$

$$\int x^2 \ln x dx = \frac{1}{3} x^3 \ln x - \frac{1}{9} x^3 + C$$

2) $\int x e^{-3x} dx$

Set $u = x$ and $dv = e^{-3x} dx$.

Then $du = dx$, $v = -\frac{1}{3} e^{-3x}$; [Formula $\int e^{kx} dx = \frac{1}{k} e^{kx} + C$]

$$\begin{aligned}
&\int x e^{-3x} dx \\
&= \int u dv = uv - \int v du \\
&= x \cdot \frac{e^{-3x}}{-3} - \int \frac{e^{-3x}}{-3} dx \\
&= -\frac{1}{3} x e^{-3x} + \frac{1}{9} e^{-3x} + C
\end{aligned}$$

3) $\int x \tan^{-1} x dx$. Here $u = \tan^{-1} x$, $dv = x dx$. Then $du = \frac{1}{1+x^2} dx$, $v = \frac{x^2}{2}$

$$\begin{aligned}
&\int x \tan^{-1} x dx \\
&= \int u dv = uv - \int v du \\
&= \tan^{-1} x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \frac{1}{1+x^2} dx \\
&= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx \\
&= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \frac{(1+x^2) - 1}{1+x^2} dx \\
&= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \left[1 - \frac{1}{1+x^2} \right] dx \\
&\int x \tan^{-1} x dx = \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + C
\end{aligned}$$

Theorem: Fundamental Theorem of Calculus - part-2

If $F(x)$ is the anti-derivative of the function $f(x)$ over the interval $[a, b]$, that is,

$$\int f(x) dx = F(x) + C, \text{ for } a \leq x \leq b,$$

then $\int_a^b f(x) dx = [F(x) + C]_a^b = F(b) - F(a)$.

Note: By the help of this Theorem, to integrate $\int_0^1 x \tan^{-1} x dx$, we can start with $\int x \tan^{-1} x dx$.

Since $\int x \tan^{-1} x dx = \frac{1}{2}x^2 \tan^{-1} x - \frac{1}{2}x + \frac{1}{2} \tan^{-1} x + C$, so

$$\int_0^1 x \tan^{-1} x dx = \left[\frac{1}{2}x^2 \tan^{-1} x - \frac{1}{2}x + \frac{1}{2} \tan^{-1} x + C \right]_0^1$$

$$= \frac{1}{2} \frac{\pi}{4} - \frac{1}{2} + \frac{1}{2} \frac{\pi}{4} + C - 0 + 0 - 0 - C = \frac{\pi}{4}$$

Note: There will remain no C in the answer of a definite integral.

4) $\int_0^1 x \cot^{-1} x dx$ Homework

5) $\int x^2 (\ln x)^2 dx$

$$u = (\ln x)^2, \quad dv = x^2 dx \Rightarrow du = \frac{2 \ln x}{x} dx, \quad v = \frac{x^3}{3}$$

Now,

$$\int x^2 (\ln x)^2 dx$$

$$= \int u dv = uv - \int v du$$

$$= (\ln x)^2 \frac{x^3}{3} - \int \frac{x^3}{3} (2 \ln x) \frac{1}{x} dx$$

$$= (\ln x)^2 \frac{x^3}{3} - \frac{2}{3} \int x^2 \ln x dx; \quad [\text{Set } u = \ln x, \quad dv = x^2 dx \Rightarrow du = \frac{1}{x} dx, \quad v = \frac{x^3}{3}]$$

$$= (\ln x)^2 \frac{x^3}{3} - \frac{2}{3} \left[\ln x \cdot \frac{x^3}{3} - \int \frac{x^3}{3} \frac{1}{x} dx \right]$$

$$= (\ln x)^2 \frac{x^3}{3} - \frac{2}{9} x^3 \ln x + \frac{2}{9} \int x^2 dx$$

$$= (\ln x)^2 \frac{x^3}{3} - \frac{2}{9} x^3 \ln x + \frac{2}{27} x^3 + C$$

Exercise: Homework

A) $\int x^3 e^{-x} dx$; Use integration by parts method 3-times OR Tabular method

B) $\int x^2 \sin x dx$; Use integration by parts method 2-times OR Tabular method

Solution : (A) $\int x^3 e^{-x} dx$; Set $u = x^3$, $dv = e^{-x} dx \Rightarrow du = 3x^2 dx$, $v = -e^{-x}$

$$\int x^3 e^{-x} dx = \int u dv = uv - \int v du$$

$$= x^3 (-e^{-x}) - \int (-e^{-x}) 3x^2 dx$$

$$= -x^3 e^{-x} + 3 \int e^{-x} x^2 dx ; [\text{Set } u = x^2, dv = e^{-x} dx \Rightarrow du = 2x dx, v = -e^{-x}]$$

$$= -x^3 e^{-x} + 3 \left[x^2 (-e^{-x}) - \int (-e^{-x}) 2x dx \right]$$

$$= -x^3 e^{-x} - 3x^2 e^{-x} + 6 \int x e^{-x} dx ; [\text{Set } = x, dv = e^{-x} dx \Rightarrow du = dx, v = -e^{-x}]$$

$$= -x^3 e^{-x} - 3x^2 e^{-x} + 6 \left[x(-e^{-x}) - \int (-e^{-x}) dx \right]$$

$$= -x^3 e^{-x} - 3x^2 e^{-x} - 6x e^{-x} + 6 \int e^{-x} dx$$

$$\int x^3 e^{-x} dx = -x^3 e^{-x} - 3x^2 e^{-x} - 6x e^{-x} - 6 e^{-x} + C$$

6) Evaluate $\int_0^{\frac{\pi}{2}} e^{2x} \sin(3x) dx$

Type of the exercise: If you apply the method integration by parts twice, then you will get back the given integral. We need to set the integral as I .

Solution: To evaluate $\int_0^{\frac{\pi}{2}} e^{2x} \sin(3x) dx$, we will start with $\int e^{2x} \sin(3x) dx$.

Set $I = \int e^{2x} \sin(3x) dx$; [(i) Set $u = \sin(3x)$, $dv = e^{2x} dx \Rightarrow du = 3 \cos(3x) dx$, $v = \frac{1}{2} e^{2x}$]

$$= \int u \, dv = uv - \int v \, du$$

$$= \sin(3x) \frac{1}{2} e^{2x} - \int \frac{1}{2} e^{2x} 3 \cos(3x) \, dx$$

$$= \frac{1}{2} e^{2x} \sin(3x) - \frac{3}{2} \int e^{2x} \cos(3x) \, dx ;$$

$$[(ii) \text{ Set } u = \cos(3x) , \, dv = e^{2x} \, dx \Rightarrow du = -3 \sin(3x) \, dx , \, v = \frac{1}{2} e^{2x}]$$

$$= \frac{1}{2} e^{2x} \sin(3x) - \frac{3}{2} \left[\cos(3x) \frac{1}{2} e^{2x} - \int \frac{1}{2} e^{2x} (-3 \sin(3x)) \, dx \right]$$

$$= \frac{1}{2} e^{2x} \sin(3x) - \frac{3}{4} \cos(3x) e^{2x} - \frac{9}{4} \int e^{2x} \sin(3x) \, dx$$

$$\Rightarrow I = \frac{1}{2} e^{2x} \sin(3x) - \frac{3}{4} \cos(3x) e^{2x} - \frac{9}{4} I$$

$$\Rightarrow I + \frac{9}{4} I = \frac{e^{2x}}{4} [2 \sin(3x) - 3 \cos(3x)]$$

$$\Rightarrow \frac{13}{4} I = \frac{e^{2x}}{4} [2 \sin(3x) - 3 \cos(3x)]$$

$$\Rightarrow I = \frac{4}{13} \cdot \frac{e^{2x}}{4} [2 \sin(3x) - 3 \cos(3x)] + C$$

$$\therefore \int e^{2x} \sin(3x) \, dx = \frac{e^{2x}}{13} [2 \sin(3x) - 3 \cos(3x)] + C$$

Now, by the Fundamental Theorem of Calculus,

$$\int_0^{\frac{\pi}{2}} e^{2x} \sin(3x) \, dx = \left[\frac{e^{2x}}{13} [2 \sin(3x) - 3 \cos(3x)] + C \right]_0^{\frac{\pi}{2}}$$

$$= \frac{e^{\pi}}{13} \left[2 \sin\left(3 \frac{\pi}{2}\right) - 3 \cos\left(3 \frac{\pi}{2}\right) \right] + C - \frac{e^0}{13} [2 \sin 0 - 3 \cos 0] - C$$

$$= \frac{e^{\pi}}{13} [2(-1) - 3(0)] - \frac{1}{13} [2(0) - 3(1)]$$

$$= \frac{e^{\pi}}{13} (-2) + \frac{1}{13} (3)$$

$$\int_0^{\frac{\pi}{2}} e^{2x} \sin(3x) \, dx = \frac{1}{13} [3 - 2e^{\pi}]$$

Section 7.3: Integration of Trigonometric Functions

Group -1: All six trigonometric functions with power 1

$$1) \int \sin x \, dx = -\cos x + C$$

$$2) \int \cos x \, dx = \sin x + C$$

$$3) \int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx = \int \frac{1}{\cos x} \sin x \, dx$$

[Set $u = \cos x \Rightarrow \frac{du}{dx} = -\sin x$, that is, $du = -\sin x \, dx$. Hence, $-du = \sin x \, dx$]

$$= \int \frac{1}{u} (-1) du = - \int \frac{1}{u} du$$

$$= -\ln |u| + C$$

$$= -\ln |\cos x| + C$$

$$= \ln |\cos x|^{-1} + C ; [n \log_b x = \log_b x^n]$$

$$= \ln |(\cos x)^{-1}| + C$$

$$\int \tan x \, dx = \ln |\sec x| + C$$

$$4) \int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx = \int \frac{1}{\sin x} \cos x \, dx = \ln |\sin x| + C$$

$$5) \int \sec x \, dx = \int \sec x \cdot 1 \, dx = \int \frac{\sec x (\tan x + \sec x)}{\tan x + \sec x} \, dx$$

$$= \int \frac{\sec x \tan x + \sec^2 x}{\tan x + \sec x} \, dx = \int \frac{\sec^2 x + \sec x \tan x}{\tan x + \sec x} \, dx ;$$

$$[\text{Set } u = \tan x + \sec x, \text{ then } \frac{du}{dx} = \sec^2 x + \sec x \tan x]$$

$$\text{Hence, } du = (\sec^2 x + \sec x \tan x) dx]$$

$$\int \sec x \, dx = \int \frac{1}{u} du = \ln |u| + C$$

$$\int \sec x \, dx = \ln |\tan x + \sec x| + C$$

$$\text{Note: (1) } \int \sec x \tan x \, dx = \sec x + C \quad \text{and} \quad \int \sec^2 x \, dx = \tan x + C$$

$$(2) \int \csc x \cot x \, dx = -\csc x + C \quad \text{and} \quad \int \csc^2 x \, dx = -\cot x + C$$

Also, $\cot^2 x + 1 = \csc^2 x \Rightarrow 1 = \csc^2 x - \cot^2 x$

6) $\int \csc x \, dx$ Homework

$$\int \csc x \, dx = \ln|\csc x - \cot x| + C, \quad \text{hint: set } u = \csc x + \cot x$$

Group -2: All six trigonometric functions with power 2

Formulas:

- (i) $\sin^2 x = \frac{1}{2}[1 - \cos(2x)]$
- (ii) $\cos^2 x = \frac{1}{2}[1 + \cos(2x)]$
- (iii) $\tan^2 x + 1 = \sec^2 x \Rightarrow \tan^2 x = \sec^2 x - 1$
- (iv) $\cot^2 x + 1 = \csc^2 x \Rightarrow \cot^2 x = \csc^2 x - 1$

$$\begin{aligned} 1) \int \sin^2 x \, dx &= \int \frac{1}{2}[1 - \cos(2x)] \, dx \\ &= \frac{1}{2} \int [1 - \cos(2x)] \, dx \\ &= \frac{1}{2} \left[x - \frac{\sin(2x)}{2} \right] + C \quad ; \quad \left[\text{Formula: } \int \cos(kx) \, dx = \frac{\sin(kx)}{k} + C \right] \\ \int \sin^2 x \, dx &= \frac{1}{2}x - \frac{1}{4}\sin(2x) + C \end{aligned}$$

$$2) \int \cos^2 x \, dx = \frac{1}{2}x + \frac{1}{4}\sin(2x) + C$$

$$3) \int \tan^2 x \, dx = \int [\sec^2 x - 1] \, dx = \tan x - x + C$$

$$4) \int \cot^2 x \, dx = \int [\csc^2 x - 1] \, dx = -\cot x - x + C$$

$$5) \int \sec^2 x \, dx = \tan x + C$$

$$6) \int \csc^2 x \, dx = -\cot x + C$$

Group -3: All six trigonometric functions with power n , $n \geq 2$.

Reduction Formulas for Integration:

$$(1) \int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$(2) \int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$$(3) \int \tan^n x \, dx = \frac{1}{n-1} \tan^{n-1} x - \int \tan^{n-2} x \, dx$$

$$(4) \int \cot^n x \, dx = -\frac{1}{n-1} \cot^{n-1} x - \int \cot^{n-2} x \, dx$$

$$(5) \int \sec^n x \, dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{(n-2)}{n-1} \int \sec^{n-2} x \, dx$$

$$(6) \int \csc^n x \, dx = -\frac{1}{n-1} \csc^{n-2} x \cot x + \frac{(n-2)}{n-1} \int \csc^{n-2} x \, dx$$

Definition: Co-functions

- (i) Sine and Cosine are co functions
- (ii) Tangent and Cotangent are co functions
- (iii) Secant and Cosecant are co functions

Evaluate $\int \sec^5 x \, dx$

Solution: we know that for $n \geq 2$,

$$\int \sec^n x \, dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{(n-2)}{n-1} \int \sec^{n-2} x \, dx \dots \dots \dots (1)$$

Here $\int \sec^5 x \, dx$; Given $n = 5$, $n - 1 = 4$, $n - 2 = 3$

$$\int \sec^5 x \, dx = \frac{1}{4} \sec^3 x \tan x + \frac{3}{4} \int \sec^3 x \, dx; \text{ here } n = 3, n - 1 = 2, n - 2 = 1$$

$$= \frac{1}{4} \sec^3 x \tan x + \frac{3}{4} \left[\frac{1}{2} \sec x \tan x + \frac{1}{2} \int \sec x \, dx \right]$$

$$= \frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \ln |\sec x + \tan x| + C$$

Homework :

1. $\int \sec^7 x \, dx + \int \sec^5 x \, dx$

2. $\int \sin^6 x \, dx + \int \sin^4 x \, dx$

3. $\int \tan^6 x \, dx + \int \tan^5 x \, dx$

Group-4: $\int \sin A \cos B \, dx$; $\int \sin A \sin B \, dx$; $\int \cos A \cos B \, dx$, where $A \neq B$.

Formulas:

- 1) $\sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$
- 2) $\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$
- 3) $\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$

Definitions

- 1) $y = \sin x$ is an odd function, that is, $\sin(-x) = -\sin x$
- 2) $y = \cos x$ is an even function, that is, $\cos(-x) = \cos x$

Here,

$$\begin{aligned}\cos(5x) \sin(2x) &= \sin(2x) \cos(5x) \\ &= \frac{1}{2} [\sin(-3x) + \sin(7x)] = \frac{1}{2} [-\sin(3x) + \sin(7x)]\end{aligned}$$

Example:1 Evaluate $\int_0^{\frac{\pi}{2}} \sin(3x) \sin(6x) \, dx$

$$\begin{aligned}\text{Solution: } &\int_0^{\frac{\pi}{2}} \sin(3x) \sin(6x) \, dx \\ &= \int_0^{\frac{\pi}{2}} \frac{1}{2} [\cos(3x - 6x) - \cos(3x + 6x)] \, dx \\ &= \int_0^{\frac{\pi}{2}} \frac{1}{2} [\cos(-3x) - \cos(9x)] \, dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} [\cos(3x) - \cos(9x)] \, dx = \frac{1}{2} \left[\frac{\sin(3x)}{3} - \frac{\sin(9x)}{9} \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{6} \sin\left(3 \cdot \frac{\pi}{2}\right) - 0 - \frac{1}{18} \sin\left(9 \cdot \frac{\pi}{2}\right) - 0 = -\frac{1}{6} - \frac{1}{18} = -\frac{4}{18} = -\frac{2}{9}\end{aligned}$$

Example: 2 Evaluate $\int_0^{\frac{\pi}{2}} \cos(3x) \cos(6x) \, dx$

$$\begin{aligned}&= \int_0^{\frac{\pi}{2}} \frac{1}{2} [\cos(3x - 6x) + \cos(3x + 6x)] \, dx \\ &= \int_0^{\frac{\pi}{2}} \frac{1}{2} [\cos(-3x) + \cos(9x)] \, dx\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \int_0^{\frac{\pi}{2}} [\cos(3x) + \cos(9x)] dx \\
&= \frac{1}{2} \left[\frac{\sin(3x)}{3} + \frac{\sin(9x)}{9} \right]_0^{\frac{\pi}{2}} \\
&= \frac{1}{6} \sin\left(3 \cdot \frac{\pi}{2}\right) - 0 + \frac{1}{18} \sin\left(9 \cdot \frac{\pi}{2}\right) - 0 \\
&= \frac{1}{6}(-1) + \frac{1}{18}(1) = -\frac{1}{6} + \frac{1}{18} \\
&= -\frac{2}{18} = -\frac{1}{9}
\end{aligned}$$

Group: 5 $\int \sin^n x \cos^m x \, dx$; here m and n are positive integers

➔ There are 3 –cases.

Case-1: When n is even and m is odd

Steps:

- 1) Split off a factor $\cos x \, dx$
- 2) Write $\cos^2 x = 1 - \sin^2 x$
- 3) Set $u = \sin x$. Then $\frac{du}{dx} = \cos x$, that is, $\cos x \, dx = du$

Example: 3 $\int \sin^4 x \cos^5 x \, dx$

Solution: Given, $\int \sin^4 x \cos^5 x \, dx$

$$\begin{aligned}
&= \int \sin^4 x \cos^4 x \cos x \, dx \\
&= \int \sin^4 x (\cos^2 x)^2 \cos x \, dx \\
&= \int \sin^4 x (1 - \sin^2 x)^2 \cdot \cos x \, dx \\
&\quad \left[\text{Set } u = \sin x. \text{ Then } \frac{du}{dx} = \cos x, \text{ that is, } \cos x \, dx = du \right] \\
&= \int u^4 (1 - u^2)^2 \, du \\
&= \int u^4 [1 - 2u^2 + u^4] \, du
\end{aligned}$$

$$\begin{aligned}
&= \int [u^4 - 2u^6 + u^8] du \\
&= \frac{1}{5}u^5 - \frac{2}{7}u^7 + \frac{1}{9}u^9 + C = \frac{1}{5}\sin^5 x - \frac{2}{7}\sin^7 x + \frac{1}{9}\sin^9 x + C
\end{aligned}$$

Case-2: When n is odd and m is even

Steps:

- 1) Split off a factor $\sin x \, dx$
- 2) Write $\sin^2 x = 1 - \cos^2 x$
- 3) Set $u = \cos x$. Then $\frac{du}{dx} = -\sin x \Rightarrow \sin x \, dx = -du$

Example:4 Evaluate $\int \sin^7 x \cos^8 x \, dx$

Solution: $\int \sin^7 x \cos^8 x \, dx = \int \sin^6 x \cos^8 x \sin x \, dx$

$$= \int (\sin^2 x)^3 \cos^8 x \sin x \, dx = \int (1 - \cos^2 x)^3 \cos^8 x \sin x \, dx$$

Set $u = \cos x$. Then $\frac{du}{dx} = -\sin x \Rightarrow \sin x \, dx = -du$

$$\begin{aligned}
\int \sin^7 x \cos^8 x \, dx &= \int (1 - u^2)^3 u^8 (-1) \, du \\
&= - \int (1 - u^2)^3 u^8 \, du \\
&= - \int [1 - 3u^2 + 3u^4 - u^6] u^8 \, du \\
&= - \int [u^8 - 3u^{10} + 3u^{12} - u^{14}] \, du \\
&= - \left[\frac{1}{9}u^9 - \frac{3}{11}u^{11} + \frac{3}{13}u^{13} - \frac{1}{15}u^{15} \right] + C \\
&= - \left[\frac{1}{9}\cos^9 x - \frac{3}{11}\cos^{11} x + \frac{3}{13}\cos^{13} x - \frac{1}{15}\cos^{15} x \right] + C
\end{aligned}$$

Case-3: When n and m both are even

Step: Write $\sin^2 x = \frac{1}{2}[1 - \cos(2x)]$ and $\cos^2 x = \frac{1}{2}[1 + \cos(2x)]$; and whenever we get $\cos^2 \theta$, we must apply the formula $\cos^2 \theta = \frac{1}{2}[1 + \cos(2\theta)]$.

Example: Evaluate $\int \sin^4 x \cos^4 x \, dx$

$$\begin{aligned} &= \int (\sin^2 x \cos^2 x)^2 \, dx \\ &= \int \left(\frac{1}{2} [1 - \cos(2x)] \right)^2 \left(\frac{1}{2} [1 + \cos(2x)] \right)^2 \, dx \\ &= \frac{1}{4} \int (1 - \cos^2(2x))^2 \, dx \\ &= \frac{1}{16} \int \left(1 - \frac{1}{2} [1 + \cos(4x)] \right)^2 \, dx \\ &= \frac{1}{16} \int \left(1 - \frac{1}{2} - \frac{1}{2} \cos(4x) \right)^2 \, dx \\ &= \frac{1}{16} \int \left(\frac{1}{2} - \frac{1}{2} \cos(4x) \right)^2 \, dx \\ &= \frac{1}{16} \cdot \frac{1}{4} \int (1 - \cos(4x))^2 \, dx \\ &= \frac{1}{64} \int [1 - 2\cos(4x) + \cos^2(4x)] \, dx \\ &= \frac{1}{64} \int [1 - 2\cos(4x)] \, dx + \frac{1}{64} \int \cos^2(4x) \, dx \\ &= \frac{1}{64} \left[x - 2 \frac{\sin(4x)}{4} \right] + \frac{1}{64} \int \frac{1}{2} [1 + \cos(8x)] \, dx \\ &= \frac{1}{64} x - \frac{1}{128} \sin(4x) + \frac{1}{64} \cdot \frac{1}{2} \left[x + \frac{\sin(8x)}{8} \right] + C \\ &= \frac{3}{128} x - \frac{1}{128} \sin(4x) + \frac{1}{1024} \sin(8x) + C \end{aligned}$$

Alternative Method:

$$\begin{aligned} \text{Evaluate } \int \sin^4 x (1 - \sin^2 x)^2 \, dx &= \int [\sin^4 x - 2\sin^6 x + \sin^8 x] \, dx \\ &= \int \sin^4 x \, dx - 2 \int \sin^6 x \, dx + \int \sin^8 x \, dx \end{aligned}$$

Homework

Evaluate $\int \sin^{11} x \cos^5 x \, dx$

*****Evaluate $\int \sin^6 x \cos^4 x \, dx = \int (\sin^4 x \cos^4 x) \sin^2 x \, dx$ Homework**

Evaluate $\int \sin^6 x \cos^5 x \, dx$

Evaluate $\int \sin^{99} x \cos^7 x \, dx$

Group: 6 $\int \tan^n x \sec^m x \, dx$; here m and n are positive integers

Group: 7 $\int \cot^n x \csc^m x \, dx$; here m and n are positive integers

Group: 6 $\int \tan^n x \sec^m x \, dx$; here m and n are positive integers

There are 3-cases in Group 6

Case-1: If m is even

Steps:

- 1) Split off the factor $\sec^2 x \, dx$; [Do not change this factor. Save it for du]
- 2) Write $\sec^2 x = 1 + \tan^2 x$
- 3) Set $u = \tan x$. Then we get $du = \sec^2 x \, dx$

Example: 5 $\int \tan^{100} x \sec^2 x \, dx = \int u^{100} \, du = \frac{1}{101} \tan^{101} x + C$

Set $u = \tan x$. Then $du = \sec^2 x \, dx$

Continue!!!

Example: 6 $\int \tan^9 x \sec^6 x \, dx$

$$= \int \tan^9 x \sec^4 x \sec^2 x \, dx$$

$$= \int \tan^9 x (\sec^2 x)^2 \sec^2 x \, dx$$

$$= \int \tan^9 x (1 + \tan^2 x)^2 \sec^2 x \, dx ; \text{ Set } u = \tan x, \text{ then } du = \sec^2 x \, dx$$

$$= \int u^9 (1 + u^2)^2 \, du$$

$$= \int u^9 [1 + 2u^2 + u^4] \, du$$

$$= \int [u^9 + 2u^{11} + u^{13}] \, du$$

$$= \frac{1}{10} u^{10} + \frac{2}{12} u^{12} + \frac{1}{14} u^{14} + C$$

$$= \frac{1}{10} \tan^{10} x + \frac{2}{12} \tan^{12} x + \frac{1}{14} \tan^{14} x + C$$

Group: 6 $\int \tan^n x \sec^m x \, dx$; here m and n are positive integers

Case-2: If n is odd

Steps:

- 1) Split off the factor $\sec x \tan x \, dx$
- 2) Write $\tan^2 x = \sec^2 x - 1$
- 3) Set $u = \sec x$. Then we get $du = \sec x \tan x \, dx$

Example: 7 $\int \tan^7 x \sec^{10} x \, dx$ **Homework**

Case-3: If n is even and m is odd

Step: Write $\tan^2 x = \sec^2 x - 1$, and then we will get sum of integrals of the form

$\int \sec^k x \, dx$ for $k \geq 2$. So, apply the reduction formula

$$\int \sec^k x \, dx = \frac{1}{k-1} \sec^{k-2} x \tan x + \frac{k-2}{k-1} \int \sec^{k-2} x \, dx \dots \dots \dots (1)$$

Example: 8 $\int \tan^4 x \sec^3 x \, dx = \int (\tan^2 x)^2 \sec^3 x \, dx$

$$= \int (\sec^2 x - 1)^2 \sec^3 x \, dx$$

$$= \int [\sec^4 x - 2 \sec^2 x + 1] \sec^3 x \, dx$$

$$= \int [\sec^7 x - 2 \sec^5 x + \sec^3 x] \, dx$$

$$= \int \sec^7 x \, dx - 2 \int \sec^5 x \, dx + \int \sec^3 x \, dx \text{ Complete !!!}$$

Group: 7 $\int \cot^n x \csc^m x \, dx$; here m and n are positive integers .

Homework: Write all the steps for all 3-cases for Group-7 and then solve following exercises

Example: 9 $\int \cot^9 x \csc^6 x \, dx$

Example: 10 $\int \cot^7 x \csc^{16} x \, dx$

Group: 7 $\int \cot^n x \csc^m x dx$; here m and n are positive integers .

Case-3: If n is even and m is odd

Step: Write $\cot^2 x = \csc^2 x - 1$, and then we will get sum of integrals of the form

$\int \csc^k x dx$ for $k \geq 2$. So, apply the reduction formula

$$(1) \quad \dots \dots \dots \int \csc^k x dx = -\frac{1}{k-1} \csc^{k-2} x \cot x + \frac{k-2}{k-1} \int \csc^{k-2} x dx$$

$$\text{Example: 11} \quad \int \cot^4 x \csc^3 x dx = \int (\cot^2 x)^2 \csc^3 x dx$$

$$= \int (\csc^2 x - 1)^2 \csc^3 x dx$$

$$= \int [\csc^4 x - 2 \csc^2 x + 1] \csc^3 x dx$$

$$= \int [\csc^7 x - 2 \csc^5 x + \csc^3 x] dx$$

$$[\int \csc^k x dx = -\frac{1}{k-1} \csc^{k-2} x \cot x + \frac{k-2}{k-1} \int \csc^{k-2} x dx]$$

$$= \int \csc^7 x dx - 2 \int \csc^5 x dx + \int \csc^3 x dx ; k = 7, \quad k - 1 = 6, k - 2 = 5$$

$$= -\frac{1}{6} \csc^5 x \cot x + \frac{5}{6} \int \csc^5 x dx - 2 \int \csc^5 x dx + \int \csc^3 x dx$$

$$= -\frac{1}{6} \csc^5 x \cot x + \left(\frac{5}{6} - 2\right) \int \csc^5 x dx + \int \csc^3 x dx$$

$$= -\frac{1}{6} \csc^5 x \cot x - \frac{7}{6} \int \csc^5 x dx + \int \csc^3 x dx$$

$$= -\frac{1}{6} \csc^5 x \cot x - \frac{7}{6} \left[-\frac{1}{4} \csc^3 x \cot x + \frac{3}{4} \int \csc^3 x dx \right] + \int \csc^3 x dx$$

... .. Continue !!

Section 7.4 Integration by Trigonometric Substitutions

Understanding:

- ➔ Trigonometric Functions [Chapter 6, MAT 116]
- ➔ Inverse Trigonometric Functions. [2 sections from chapter 8, different book]

TRIGONOMETRIC SUBSTITUTIONS: $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2}$

Formulas:

$$1 - \sin^2 x = \cos^2 x ; \quad 1 + \tan^2 x = \sec^2 x ; \quad \sec^2 x - 1 = \tan^2 x$$

We need this method to integrate type (b) integrals.

Examples: (1) a) $\int x(4 - x^2)^{44} dx$ and b) $\int (4 - x^2)^{44} dx$

$$(2) a) \int 2x\sqrt{1 - x^2} dx \quad \text{and} \quad b) \int \sqrt{1 - x^2} dx$$

Note: $\sqrt{9} = 3$ and $-\sqrt{9} = -3$ [Solve $x^2 = 9 \rightarrow x = \pm\sqrt{9}$, that is, $x = +\sqrt{9}$, $x = -\sqrt{9}$]

Definition: For any real number x , $\sqrt{x^2} = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

Notes:

- (1) $y = \sqrt{x}$: This is called **positive square root**, and we only get non-negative number y from this equation. Here, $y = 0$ if $x = 0$. Example: $\sqrt{4} = 2$.
- (2) $y = -\sqrt{x}$: This is called **negative square root**, and we only get non-positive number y from this equation. Here, $y = 0$ if $x = 0$. Example: $-\sqrt{4} = -2$.

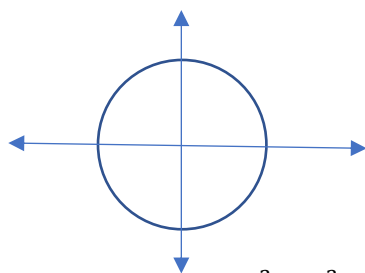
Recall: To solve $x^2 = 9 \Rightarrow x = \pm\sqrt{9} \Rightarrow x = \pm 3$. That is, $x = 3$ or $x = -3$

There are 3-cases

TRIGONOMETRIC SUBSTITUTIONS

EXPRESSION IN THE INTEGRAND	SUBSTITUTION	RESTRICTION ON θ	SIMPLIFICATION
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$-\pi/2 \leq \theta \leq \pi/2$	$a^2 - x^2 = a^2 - a^2 \sin^2 \theta = a^2 \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$-\pi/2 < \theta < \pi/2$	$a^2 + x^2 = a^2 + a^2 \tan^2 \theta = a^2 \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\begin{cases} 0 \leq \theta < \pi/2 & (\text{if } x \geq a) \\ \pi/2 < \theta \leq \pi & (\text{if } x \leq -a) \end{cases}$	$x^2 - a^2 = a^2 \sec^2 \theta - a^2 = a^2 \tan^2 \theta$

Recall: Consider the circle of radius r and with center at the origin.



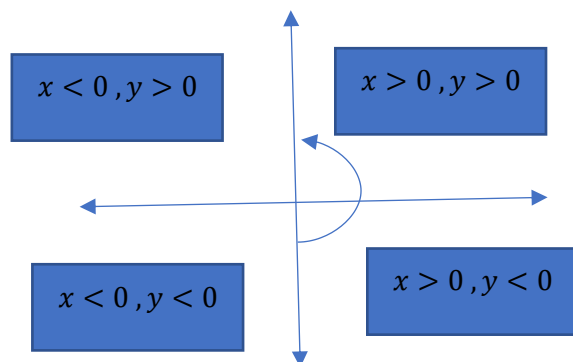
For any point (x, y) on the circle, we get $x^2 + y^2 = r^2$, and also, in polar coordinates,

$$x = r \cos \theta, \quad y = r \sin \theta.$$

In particular, for any point (x, y) on the unit circle, we get $x^2 + y^2 = 1$, and also, in polar coordinates,

$$x = \cos \theta, \quad y = \sin \theta.$$

Remember that $(x, y) = (\cos \theta, \sin \theta)$.



Case: 1 $\sqrt{a^2 - x^2}$; $a > 0$

Set $x = a \sin \theta$. Then $\frac{dx}{d\theta} = a \cos \theta \Rightarrow dx = a \cos \theta d\theta$

$$\text{Also, } \sqrt{a^2 - x^2} = \sqrt{a^2 - (a \sin \theta)^2}$$

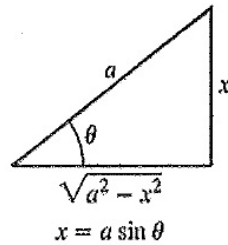
$$= \sqrt{a^2 - a^2 \sin^2 \theta} = \sqrt{a^2(1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta} = \sqrt{a^2} \sqrt{\cos^2 \theta} = |a| |\cos \theta|$$

$$= a |\cos \theta| = a \cos \theta \text{ for } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

That is, for case -I: $\sqrt{a^2 - x^2} = a \cos \theta$; when $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

Again, $x = a \sin \theta \Rightarrow \frac{x}{a} = \sin \theta$. Hence $\theta = \sin^{-1}\left(\frac{x}{a}\right)$

Now, $\sin \theta = \frac{x}{a} = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{x \text{ is the length of opposite}}{a \text{ is the length of hypotenuse}}$



$$\cos \theta = \frac{\sqrt{a^2 - x^2}}{a}, \sec \theta = \frac{a}{\sqrt{a^2 - x^2}}, \tan \theta = \frac{x}{\sqrt{a^2 - x^2}}, \cot \theta = \frac{\sqrt{a^2 - x^2}}{x}, \csc \theta = \frac{a}{x}$$

Case: 2 $\sqrt{a^2 + x^2}$; $a > 0$

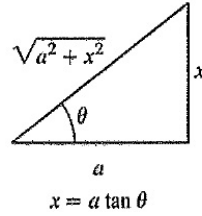
Set $x = a \tan \theta$. Then $\frac{dx}{d\theta} = a \sec^2 \theta \Rightarrow dx = a \sec^2 \theta d\theta$

$$\sqrt{a^2 + x^2} = \sqrt{a^2 + (a \tan \theta)^2} = \sqrt{a^2(1 + \tan^2 \theta)} = \sqrt{a^2 \sec^2 \theta} = a \sqrt{\sec^2 \theta} = a |\sec \theta|$$

$\sqrt{a^2 + x^2} = a \sec \theta$; when $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ [$\sec \theta = \frac{1}{\cos \theta} \rightarrow \theta \neq -\frac{\pi}{2}, \frac{\pi}{2}$]

Again, $x = a \tan \theta \Rightarrow \frac{x}{a} = \tan \theta$. Hence $\theta = \tan^{-1}\left(\frac{x}{a}\right)$

Now, $\tan \theta = \frac{x}{a} = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{x \text{ is the opposite}}{a \text{ is the Adjacent}}$



$$\cos \theta = \frac{a}{\sqrt{a^2+x^2}}, \sec \theta = \frac{\sqrt{a^2+x^2}}{a}, \sin \theta = \frac{x}{\sqrt{a^2+x^2}}, \cot \theta = \frac{a}{x}, \csc \theta = \frac{\sqrt{a^2+x^2}}{x}$$

Case: 3 $\sqrt{x^2 - a^2}$; $a > 0$

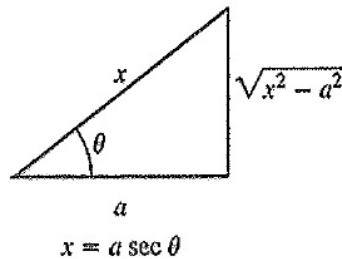
Set $x = a \sec \theta$. Then $\frac{dx}{d\theta} = a \sec \theta \tan \theta \Rightarrow dx = a \sec \theta \tan \theta d\theta$

$$\sqrt{x^2 - a^2} = \sqrt{(a \sec \theta)^2 - a^2} = \sqrt{a^2(\sec^2 \theta - 1)} = \sqrt{a^2 \tan^2 \theta} = a \sqrt{\tan^2 \theta} = a |\tan \theta|$$

$$\sqrt{x^2 - a^2} = a \tan \theta ; \text{ when } 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2} \left[\tan \theta = \frac{\sin \theta}{\cos \theta} \rightarrow \theta \neq \frac{\pi}{2}, \frac{3\pi}{2} \right]$$

Again, $x = a \sec \theta \Rightarrow \frac{x}{a} = \sec \theta$. Hence $\theta = \sec^{-1} \left(\frac{x}{a} \right)$

Now, $\sec \theta = \frac{x}{a} = \frac{\text{Hypotenuse}}{\text{Adjacent}} = \frac{x \text{ is the Hypotenuse}}{a \text{ is the Adjacent}}$



$$\cos \theta = \frac{a}{x}, \cot \theta = \frac{a}{\sqrt{x^2-a^2}}, \sin \theta = \frac{\sqrt{x^2-a^2}}{x}, \tan \theta = \frac{\sqrt{x^2-a^2}}{a}, \csc \theta = \frac{x}{\sqrt{x^2-a^2}}$$

Note: To evaluate an indefinite integral with this method, there are 3-steps.

- ➔ Use substitution and simplify, resulting an integral of trigonometric functions
- ➔ Evaluate the integral
- ➔ Re-substitute using a right triangle.

And for a definite integral, be very careful when you simplify the absolute value on the given closed interval.

Examples

$$1. \int_{\frac{2\pi}{3}}^{\pi} |\tan \theta| d\theta = \int_{\frac{2\pi}{3}}^{\pi} (-\tan \theta) d\theta; \text{ Here } \frac{2\pi}{3} \leq \theta \leq \pi, \text{ so } \tan \theta \leq 0.$$

Note: $[x = -2, |x| = |-2| = 2 = -(-2) = -x]$

$$\begin{aligned} 2. \int_0^{\pi} |\cos \theta| d\theta &= \int_0^{\frac{\pi}{2}} |\cos \theta| d\theta + \int_{\frac{\pi}{2}}^{\pi} |\cos \theta| d\theta = \int_0^{\frac{\pi}{2}} \cos \theta d\theta + \int_{\frac{\pi}{2}}^{\pi} (-\cos \theta) d\theta \\ &= \int_0^{\frac{\pi}{2}} \cos \theta d\theta - \int_{\frac{\pi}{2}}^{\pi} \cos \theta d\theta = 2 \end{aligned}$$

Section 7.4 Integration by Trigonometric Substitutions

EXERCISES

TRIGONOMETRIC SUBSTITUTIONS: $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2}$

Exercise: 1 (a) $\int \frac{\sqrt{1+t^2}}{t} dt$

Set $t = \tan \theta$. Then $\frac{dt}{d\theta} = \sec^2 \theta$, that is, $dt = \sec^2 \theta d\theta$

Now,

$$\int \frac{\sqrt{1+t^2}}{t} dt = \int \frac{\sqrt{1+\tan^2 \theta}}{\tan \theta} \sec^2 \theta d\theta = \int \frac{\sqrt{\sec^2 \theta}}{\tan \theta} \sec^2 \theta d\theta$$

$$= \int \frac{|\sec \theta|}{\tan \theta} \sec^2 \theta d\theta$$

$$= \int \frac{\sec \theta}{\tan \theta} \sec^2 \theta d\theta; \text{ when } -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$= \int \frac{\sec^3 \theta}{\tan \theta} d\theta = \int \frac{\frac{1}{\cos^3 \theta}}{\frac{\sin \theta}{\cos \theta}} d\theta$$

$$= \int \frac{1}{\cos^2 \theta \sin \theta} d\theta$$

$$= \int \sec^2 \theta \csc \theta d\theta$$

$$= \int (1 + \tan^2 \theta) \csc \theta d\theta$$

$$= \int (\csc \theta + \tan^2 \theta \csc \theta) d\theta$$

$$= \int \csc \theta d\theta + \int \tan^2 \theta \csc \theta d\theta$$

$$\begin{aligned}
&= \int \csc \theta \, d\theta + \int \frac{\sin \theta}{\cos^2 \theta} \, d\theta \quad ; \quad \text{Set } u = \cos \theta, \quad \text{then } -du = \sin \theta \, d\theta \\
&= \int \csc \theta \, d\theta + \int \frac{1}{u^2} (-1) \, du \\
&= \ln |\csc \theta - \cot \theta| - \int u^{-2} \, du \\
&= \ln |\csc \theta - \cot \theta| - \frac{u^{-2+1}}{-2+1} + C \\
&= \ln |\csc \theta - \cot \theta| + \frac{1}{u} + C \\
&= \ln |\csc \theta - \cot \theta| + \frac{1}{\cos \theta} + C \\
&= \ln |\csc \theta - \cot \theta| + \sec \theta + C \\
&= \ln \left| \frac{\sqrt{1+t^2}}{t} - \frac{1}{t} \right| + \sqrt{1+t^2} + C ;
\end{aligned}$$

[if $t = \tan \theta$, then Opposite = t , Adjacent = 1, Hypotenuse = $\sqrt{1+t^2}$]

$$\therefore \int \frac{\sqrt{1+t^2}}{t} \, dt = \ln \left| \frac{\sqrt{1+t^2}-1}{t} \right| + \sqrt{1+t^2} + C$$

$$(b) \int \frac{1}{x^2 \sqrt{x^2+5}} \, dx$$

$$\text{Set } x = \sqrt{5} \tan \theta. \quad dx = \sqrt{5} \sec^2 \theta \, d\theta$$

$$\begin{aligned}
&\int \frac{1}{x^2 \sqrt{x^2+5}} \, dx \\
&= \int \frac{1}{(\sqrt{5} \tan \theta)^2 \sqrt{(\sqrt{5} \tan \theta)^2+5}} \sqrt{5} \sec^2 \theta \, d\theta
\end{aligned}$$

$$= \int \frac{\sqrt{5} \sec^2 \theta}{5 \tan^2 \theta \sqrt{5(\tan^2 \theta+1)}} \, d\theta$$

$$= \int \frac{\sqrt{5} \sec^2 \theta}{5 \tan^2 \theta \sqrt{5 \sec^2 \theta}} \, d\theta$$

$$= \int \frac{\sqrt{5} \sec^2 \theta}{5 \tan^2 \theta |\sqrt{5} \sec \theta|} \, d\theta$$

$$= \int \frac{\sqrt{5} \sec^2 \theta}{5 \tan^2 \theta \sqrt{5} \sec \theta} \, d\theta \quad ; \quad \text{when } -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$= \frac{1}{5} \int \frac{\sec \theta}{\tan^2 \theta} d\theta$$

$$= \frac{1}{5} \int \frac{\frac{1}{\cos \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta}} d\theta$$

$$= \frac{1}{5} \int \frac{\cos \theta}{\sin^2 \theta} d\theta \quad ; \quad \text{set } u = \sin \theta, \quad du = \cos \theta d\theta$$

$$= \frac{1}{5} \int \frac{1}{u^2} du = \frac{1}{5} \int u^{-2} du = \frac{1}{5} \frac{u^{-2+1}}{-2+1} = -\frac{1}{5} \frac{1}{u} + C$$

$$= -\frac{1}{5 \sin \theta} + C = -\frac{\sqrt{x^2+5}}{5x} + C$$

$$\left[x = \sqrt{5} \tan \theta \Rightarrow \tan \theta = \frac{x}{\sqrt{5}} \quad . \quad \text{Hence, } \sin \theta = \frac{x}{\sqrt{x^2+5}} \quad , \quad \text{that is, } \frac{1}{\sin \theta} = \frac{\sqrt{x^2+5}}{x} \right]$$

$$(c) \int \frac{1}{x^2 \sqrt{x^2-4}} dx \quad \text{Set } x = 2 \sec \theta . \text{ Then } dx = 2 \sec \theta \tan \theta d\theta$$

Now,

$$\int \frac{1}{x^2 \sqrt{x^2-4}} dx$$

$$= \int \frac{1}{(2 \sec \theta)^2 \sqrt{(2 \sec \theta)^2 - 4}} 2 \sec \theta \tan \theta d\theta$$

$$= \int \frac{2 \sec \theta \tan \theta}{4 \sec^2 \theta \sqrt{4 \sec^2 \theta - 4}} d\theta$$

$$= \int \frac{\tan \theta}{2 \sec \theta \sqrt{4(\sec^2 \theta - 1)}} d\theta$$

$$= \int \frac{\tan \theta}{4 \sec \theta \sqrt{\tan^2 \theta}} d\theta$$

$$= \int \frac{\tan \theta}{4 \sec \theta |\tan \theta|} d\theta$$

$$= \int \frac{\tan \theta}{4 \sec \theta \tan \theta} d\theta ; \text{ for } 0 \leq \theta < \frac{\pi}{2} \quad \text{or } \pi \leq \theta < \frac{3\pi}{2}.$$

$$= \frac{1}{4} \int \frac{1}{\sec \theta} d\theta$$

$$= \frac{1}{4} \int \cos \theta d\theta$$

$$= \frac{1}{4} \sin \theta + C ; \quad x = 2 \sec \theta \rightarrow \sec \theta = \frac{x}{2} = \frac{\text{hyp}}{\text{adj}} \rightarrow \text{opp.} = \sqrt{x^2 - 4}$$

$$= \frac{1}{4} \frac{\sqrt{x^2 - 4}}{x} + C = \frac{\sqrt{x^2 - 4}}{4x} + C$$

Exercise: 2

$$\int \frac{1}{(1 - x^2)^{\frac{3}{2}}} dx.$$

Solution: Set $x = \sin \theta$. Then, $dx = \cos \theta d\theta$. Now,

$$\int \frac{1}{(1 - x^2)^{\frac{3}{2}}} dx = \int \frac{1}{(1 - (\sin \theta)^2)^{\frac{3}{2}}} \cos \theta d\theta$$

$$= \int \frac{1}{(1 - \sin^2 \theta)^{\frac{3}{2}}} \cos \theta d\theta = \int \frac{1}{(\cos^2 \theta)^{\frac{3}{2}}} \cos \theta d\theta$$

$$= \int \frac{1}{(\cos^2 \theta)^{\frac{3}{2}}} \cos \theta d\theta = \int \frac{1}{(\sqrt{\cos^2 \theta})^3} \cos \theta d\theta$$

$$= \int \frac{1}{(|\cos \theta|)^3} \cos \theta d\theta$$

$$= \int \frac{1}{\cos^3 \theta} \cos \theta d\theta ; \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

$$= \int \sec^2 \theta dx$$

$$= \tan \theta + C = \frac{x}{\sqrt{1-x^2}} + C$$

Exercise: 3

Evaluate

$$\int_1^{\sqrt{2}} \frac{1}{x^2 \sqrt{4-x^2}} dx.$$

$$\text{Given } \int_1^{\sqrt{2}} \frac{1}{x^2 \sqrt{4-x^2}} dx = \int_1^{\sqrt{2}} \frac{1}{x^2 \sqrt{2^2-x^2}} dx$$

Set $x = 2 \sin y$. Then $dx = 2 \cos y \, dy$. Also,

x	1	$\sqrt{2}$
y	$\frac{\pi}{6}$	$\frac{\pi}{4}$

If $x = 1$, then $1 = 2 \sin y \Rightarrow \sin y = \frac{1}{2}$, that is, $y = \frac{\pi}{6}$ and if $x = \sqrt{2}$, then $y = \frac{\pi}{4}$

$$\text{Hence, } \int_1^{\sqrt{2}} \frac{dx}{x^2 \sqrt{4-x^2}} = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{(2 \sin y)^2 \sqrt{4-4 \sin^2 y}} 2 \cos y \, dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{4 \sin^2 y \sqrt{4 \cos^2 y}} 2 \cos y \, dy = \frac{1}{4} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2 y \sqrt{(2 \cos y)^2}} 2 \cos y \, dy$$

$$= \frac{1}{4} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2 y |2 \cos y|} 2 \cos y \, dy = \frac{1}{4} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2 y 2 \cos y} 2 \cos y \, dy$$

$$= \frac{1}{4} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin^2 y} \, dy = \frac{1}{4} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \csc^2 y \, dy = \frac{1}{4} [-\cot y]_{\frac{\pi}{6}}^{\frac{\pi}{4}} = \frac{1}{4} \left[-\cot \frac{\pi}{4} + \cot \frac{\pi}{6} \right] = \frac{-1 + \sqrt{3}}{4}$$

$$\therefore \int_1^{\sqrt{2}} \frac{dx}{x^2 \sqrt{4-x^2}} = \frac{\sqrt{3}-1}{4}$$

Exercise: 4 $\int \frac{x^2}{(x^2-1)^{\frac{3}{2}}} dx$ Homework

Section 7.5 Integration by Partial Fractions

How to integrate following functions?

$$\int \frac{x^2 - 2x - 1}{x^2 - 1} dx, \quad \int \frac{x^3 - 2x - 1}{x^2 - 1} dx, \quad \int \frac{x^2 - 2x - 1}{x^3 - x} dx.$$

➔ Partial Fraction is based on Logarithmic Derivative.

Definition: Logarithmic Derivative

$$\frac{d}{dx} [\ln(f(x))] = \frac{1}{f(x)} f'(x) = \frac{f'(x)}{f(x)} = \frac{\text{Derivative of the Denominator}}{\text{Denominator}}.$$

The fraction of the type $\frac{f'(x)}{f(x)}$ is called the logarithmic derivative.

Here, $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$; $\int \frac{\cos x}{\sin x} dx = \ln|\sin x| + C.$

$$\int \frac{x}{x^2 + 5} dx = \frac{1}{2} \int \frac{2x}{x^2 + 5} dx = \frac{1}{2} \ln(x^2 + 5) + C$$

Integration by Partial Fractions:

Method: Write the given fraction as the sum of partial fractions.

- ➔ First and foremost, divide the given fraction, using long division, if it is not a proper fraction.
- ➔ Write the given (proper) fraction as a sum of partial fractions taking each factor of the denominator as a denominator of a partial fraction.
- ➔ If the power of the denominator is 1, then write a partial fraction in a way so that the numerator is a polynomial of degree n , **in general form**, where the derivative of the denominator is a polynomial of degree n .

For example,

$$1. \frac{x^2-2x-1}{x^3+x} = \frac{x^2-2x-1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

- ➔ Observe that, $\frac{Ax^3+Bx^2+Cx+D}{x^4} = \frac{Ax^3}{x^4} + \frac{Bx^2}{x^4} + \frac{Cx}{x^4} + \frac{D}{x^4} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x^4}$. In the same manner, we are going to set the partial fractions for the following fractions:

Examples:

$$(i) \frac{x^2-2x-1}{x(x+1)^3} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3}$$

$$(ii) \frac{x^2-2x-1}{x^2(x^2+1)^3} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1} + \frac{Ex+F}{(x^2+1)^2} + \frac{Gx+H}{(x^2+1)^3}$$

Example 0 Write partial fractions of the following fractions

$$(1) \frac{x^2-2x-1}{x^4-x^3} = \frac{x^2-2x-1}{x^3(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x-1}$$

$$(2) \frac{x^2-2x-1}{x^3-x} = \frac{x^2-2x-1}{x(x^2-1)} = \frac{x^2-2x-1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$$

$$(3) \frac{x^2-2x-1}{x^4-6x^3+8x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2} + \frac{D}{x-4}$$

Note: After you set up the partial fractions, you must multiply both sides of the equation by the least common denominator. Then there are two different ways to find values of those constants used in the setup.

1) **By plugging values for x**

2) By equating coefficients from both sides.

Power of $x^2 + 1$ is 1, because $x^2 + 1 = (x^2 + 1)^1$

Example :1

(a) Evaluate $\int \frac{x^2-2x-1}{x^3-x} dx$.

(b) Evaluate $\int \frac{2x-1}{x^3-x} dx$.

(c) Evaluate $\int \frac{x^2-1}{x^3-x} dx$.

Solution [Example: 1, Part (a)] Evaluate

$$\int \frac{x^2 - 2x - 1}{x^3 - x} dx.$$

Solution: Let's start with the fraction.

$$\frac{x^2 - 2x - 1}{x^3 - x} = \frac{x^2 - 2x - 1}{x(x-1)(x+1)}$$

Set $\frac{x^2-2x-1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} \dots \dots \dots (1)$; [Rational Function with D= all reals, except $-1, 0, 1$]

Multiply the both sides of the equation above by $x(x-1)(x+1)$:

$$x^2 - 2x - 1 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1) \dots \dots \dots (2) ; [\text{Polynomial, D= all reals}]$$

[(2) gives us a polynomial, and the domain of a polynomial is all real numbers. So, we can plug in any real value for x in (2)]

If $x = 0$: $-1 = A(-1)(1) \Rightarrow -A = -1 \therefore A = 1$.

If $x = 1$: $1^2 - 2(1) - 1 = B(1)(2) \Rightarrow 2B = -2 \therefore B = -1$.

If $x = -1$: $(-1)^2 - 2(-1) - 1 = C(-1)(-2) \Rightarrow 2C = 2 \therefore C = 1$.

Now, from equation (1):

$$\frac{x^2-2x-1}{x^3-x} = \frac{x^2-2x-1}{x(x-1)(x+1)} = \frac{1}{x} + \frac{-1}{x-1} + \frac{1}{x+1}$$

$$\int \frac{x^2 - 2x - 1}{x^3 - x} dx = \int \left[\frac{1}{x} + \frac{-1}{x-1} + \frac{1}{x+1} \right] dx$$

$$= \int \left[\frac{1}{x} - \frac{1}{x-1} + \frac{1}{x+1} \right] dx$$

$$= \ln|x| - \ln|x-1| + \ln|x+1| + C$$

$$= \ln|x| + \ln|x+1| - \ln|x-1| + C$$

$$= \ln|x(x+1)| - \ln|x-1| + C$$

$$\int \frac{x^2 - 2x - 1}{x^3 - x} dx = \ln \frac{|x(x+1)|}{|x-1|} + C$$

Example: 2 Evaluate $\int \frac{x^3-2x-1}{x^2-1} dx$

$$\left[\frac{\text{Dividend}}{\text{divisor}} = \text{Quotient} + \frac{\text{Remainder}}{\text{Divisor}} \right]$$

$$\begin{array}{r} x^2 - 1 \overline{) x^3 - 2x - 1} \\ \underline{-(x^3 - x)} \\ -x - 1 \end{array}$$

Given Fraction $\frac{x^3-2x-1}{x^2-1} = x + \frac{-x-1}{x^2-1} = x - \frac{(x+1)}{(x-1)(x+1)} = x - \frac{1}{x-1}$

Here $\int \frac{x^3-2x-1}{x^2-1} = \int \left[x - \frac{1}{x-1} \right] dx = \frac{x^2}{2} - \ln|x-1| + C$

Understanding: Since $\frac{d}{dx}(x^4) = 4x^3$, the derivative of x^4 is a polynomial of degree 3.

Also, general form of a polynomial of degree 3 is given by $ax^3 + bx^2 + cx + d$

$$\frac{Ax^3+Bx^2+Cx+D}{[x]^4} = \frac{A}{x^1} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x^4}$$

$$\frac{2x+1}{(x^2+4)^4} = \frac{Ax+B}{x^2+4} + \frac{Cx+D}{(x^2+4)^2} + \frac{Ex+F}{(x^2+4)^3} + \frac{Gx+H}{(x^2+4)^4}$$

Example: 3 (a) Evaluate $\int \frac{3x^2-x+1}{x^3-x^2} dx$

Solution: The given fraction is $\frac{3x^2-x+1}{x^3-x^2} = \frac{3x^2-x+1}{x^2(x-1)}$

Set $\frac{3x^2-x+1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} \dots \dots \dots (1)$

Multiply the both sides of the equation above by $x^2(x-1)$:

$$3x^2 - x + 1 = Ax(x-1) + B(x-1) + Cx^2 \dots \dots \dots (2)$$

[(2) gives us a polynomial, and the domain of a polynomial is all real numbers. So, we can plug in any real value for x in (2)]

If $x = 0$: $1 = 0 + B(-1) + 0 \Rightarrow -B = 1 \therefore B = -1$

If $x = 1$: $3 = C(1) \therefore C = 3$

If $x = -1$: $5 = A(-1)(-2) + B(-2) + C$

$$\Rightarrow 5 = 2A - 2B + C$$

$$\Rightarrow 5 = 2A - 2(-1) + 3$$

$$\Rightarrow 2A = 5 - 5$$

$$\therefore A = 0$$

$$\frac{3x^2 - x + 1}{x^3 - x^2} = \frac{3x^2 - x + 1}{x^2(x - 1)} = \frac{0}{x} + \frac{-1}{x^2} + \frac{3}{x - 1}$$

$$\int \frac{3x^2 - x + 1}{x^3 - x^2} dx = \int \left[\frac{-1}{x^2} + 3 \frac{1}{x - 1} \right] dx$$

$$= \int \left[-x^{-2} + 3 \frac{1}{x - 1} \right] dx = \frac{1}{x} + 3 \ln|x - 1| + C$$

$$\int \frac{3x^2 - x + 1}{x^3 - x^2} dx = \frac{1}{x} + 3 \ln|x - 1| + C.$$

Example: 3 (b) Evaluate

$$\int \frac{3x^2 - x + 1}{x^4 - x^3} dx.$$

$$\frac{3x^2 - x + 1}{x^4 - x^3} = \frac{3x^2 - x + 1}{x^3(x - 1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x - 1} \dots \dots \dots (1)$$

$$\text{That is, } 3x^2 - x + 1 = A x^2(x - 1) + Bx(x - 1) + C(x - 1) + Dx^3 \dots \dots \dots (1)$$

$$\text{If } x = 0: 1 = -C, \quad \therefore C = -1.$$

$$\text{If } x = 1: 3 = D, \quad \therefore D = 3.$$

$$\text{If } x = -1:$$

$$5 = -2A + 2B - 2C - D$$

$$\Rightarrow -2A + 2B - 2(-1) - 3 = 5$$

$$\Rightarrow -2A + 2B = 6$$

$$\therefore A - B = -3 \dots \dots \dots (3)$$

$$\text{If } x = 2:$$

$$11 = 4A + 2B + C + 8D$$

$$\Rightarrow 4A + 2B - 1 + 24 = 11$$

$$\Rightarrow 4A + 2B = -12$$

$$\therefore 2A + B = -6 \dots \dots (4)$$

Therefore, $A = -3$, $B = 0$

$$\frac{3x^2 - x + 1}{x^4 - x^3} = \frac{-3}{x} + \frac{-1}{x^3} + \frac{3}{x-1} = -3 \frac{1}{x} - x^{-3} + 3 \frac{1}{x-1}$$

$$\int \frac{3x^2 - x + 1}{x^4 - x^3} dx = -3 \ln|x| + \frac{1}{2x^2} + 3 \ln|x-1| + C = \frac{1}{2x^2} - 3 \ln \left| \frac{x}{x-1} \right| + C.$$

Example: 4 $\int \frac{3x^2 - x + 1}{x^3 + x} dx$ Homework

Solution: $\frac{3x^2 - x + 1}{x^3 + x} = \frac{3x^2 - x + 1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} \dots \dots (1)$

$$3x^2 - x + 1 = A(x^2 + 1) + (Bx + C)x \dots \dots (2)$$

Set $x = 0, 1, -1$ in (2)

Please complete!

Example: 5

$$\int \frac{5 + 2 \ln x}{x(1 + \ln x)^2} dx$$

Given $\int \frac{5 + 2 \ln x}{x(1 + \ln x)^2} dx = \int \frac{2(1 + \ln x) + 3}{x(1 + \ln x)^2} dx = \int \frac{2(1 + \ln x) + 3}{(1 + \ln x)^2} \frac{1}{x} dx$

$$= \int \frac{2u + 3}{u^2} du \quad ; \quad \text{Set } u = 1 + \ln x, \text{ then } du = \frac{1}{x} dx.$$

That is, $\int \frac{5 + 2 \ln x}{x(1 + \ln x)^2} dx = \int \frac{2u + 3}{u^2} du = \int \left[\frac{2u}{u^2} + \frac{3}{u^2} \right] du$

$$= \int \left[2 \frac{1}{u} + 3u^{-2} \right] du = 2 \ln|1 + \ln x| - \frac{3}{1 + \ln x} + C$$

Example: 6

$$\int \frac{7 \cos \theta}{\sin^2 \theta + 5 \sin \theta - 6} d\theta .$$

Solution:

Given

$$\int \frac{7 \cos \theta}{\sin^2 \theta + 5 \sin \theta - 6} d\theta .$$

Set $u = \sin \theta$, $du = \cos \theta d\theta$

$$\text{Then } \int \frac{7 \cos \theta}{\sin^2 \theta + 5 \sin \theta - 6} d\theta = \int \frac{7}{\sin^2 \theta + 5 \sin \theta - 6} \cos \theta d\theta = 7 \int \frac{1}{u^2 + 5u - 6} du \quad \text{Complete !!}$$

Example: 7 $\int \frac{3x^4 - x^2 + 1}{x^3 + x} dx$ Homework.

$$\begin{array}{r} x^3 + x \quad \overline{) \quad \begin{array}{r} 3x^4 - x^2 + 1 \\ 3x^4 + 3x^2 \\ \hline -4x^2 + 1 \end{array}} \quad 3x \end{array}$$

Note that $(3x^4 - x^2 + 1) - (3x^4 + 3x^2) = -4x^2 + 1$

$$\frac{3x^4 - x^2 + 1}{x^3 + x} = 3x + \frac{-4x^2 + 1}{x^3 + x}$$

Find the partial fraction of $\frac{-4x^2 + 1}{x^3 + x}$.

$$\int \frac{3x^4 - x^2 + 1}{x^3 + x} dx = \int \left[3x + \frac{-4x^2 + 1}{x^3 + x} \right] dx = \int 3x dx + \int \frac{-4x^2 + 1}{x^3 + x} dx \dots \dots \dots (i)$$

$$\frac{-4x^2 + 1}{x^3 + x} = \frac{-4x^2 + 1}{x(x^2 + 1)}$$

$$\text{Set } \frac{-4x^2 + 1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} \dots \dots \dots (1)$$

Then $-4x^2 + 1 = A(x^2 + 1) + (Bx + C)x \dots\dots\dots (2)$

Set $x = 0, \quad 1, -1$

Finally, $\int \frac{3x^4 - x^2 + 1}{x^3 + x} dx = \int \left[3x + \frac{A}{x} + \frac{Bx + C}{x^2 + 1} \right] dx$

Please Complete!

7.8: Improper Integrals

Proper Integrals: If $y = f(x)$ is continuous on some closed interval $[a, b]$, then $\int_a^b f(x) dx$ is called proper integral. [f is bounded on the interval].

Improper integrals:

(1) Improper integrals with **infinite intervals** of integration:

$$\int_1^{\infty} \frac{1}{x^2} dx, \quad \int_{-\infty}^1 e^x dx, \quad \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$$

(2) Improper integrals with **infinite discontinuities** in the intervals of integration:

$$\int_{-1}^1 \frac{1}{x^2} dx, \quad \int_1^2 \frac{1}{x-1} dx, \quad \int_0^{\pi} \tan x dx$$

Fixing discontinuity:

$$[-1, 1], x \neq 0 \Rightarrow [-1, 0) \cup (0, 1]$$
$$[1, 2], x \neq 1 \Rightarrow (1, 2]$$

(3) Improper integrals with infinite discontinuities and infinite intervals of integration:

$$\int_{-\infty}^{\infty} \frac{1}{x^2} dx, \quad \int_{-\infty}^0 \frac{1}{\sqrt{x}} dx, \quad \int_{-\infty}^{\infty} \frac{1}{x^2-9} dx,$$

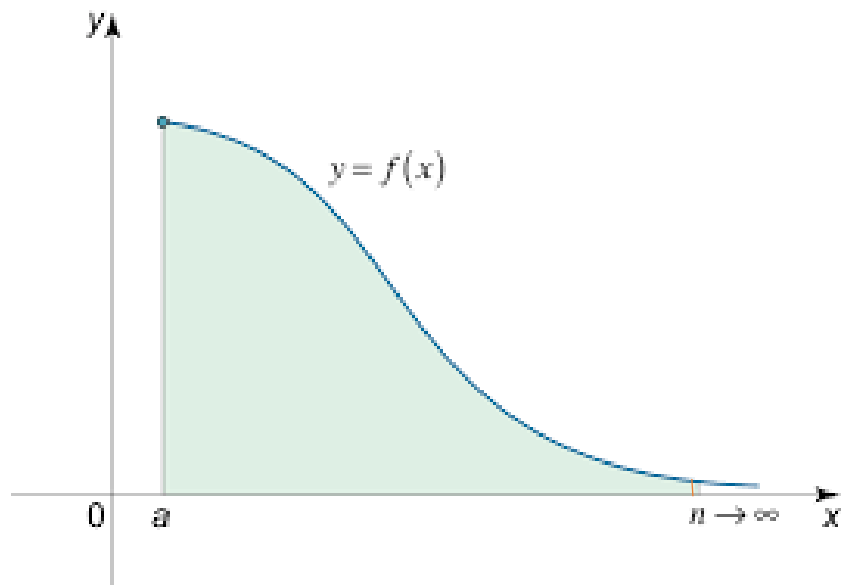
$$(-2, 3] = \lim_{a \rightarrow -2^+} [a, 3], \quad a > -2.$$

$$[-1, 0) = \lim_{b \rightarrow 0^-} [-1, b], \quad b < 0$$

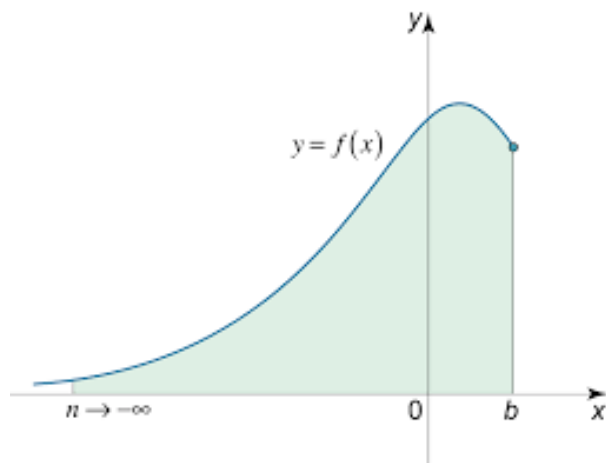
Lecture-1: Improper integrals with **infinite intervals of integration:**

To solve an improper integral with infinite interval, we need to use a finite interval with the help of limit concept. For examples,

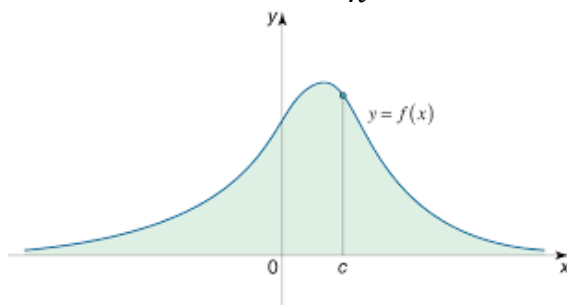
$$(a) \quad [a, \infty) = \lim_{n \rightarrow \infty} [a, n]$$

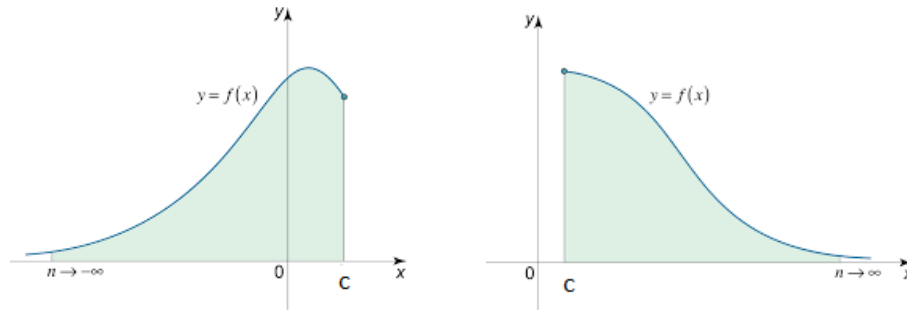


$$(b) \quad (-\infty, b] = \lim_{n \rightarrow -\infty} [n, b]$$



$$(c) \quad (-\infty, \infty) = (-\infty, c] \cup [c, \infty) = \lim_{n \rightarrow -\infty} [n, c] \cup \lim_{n \rightarrow \infty} [c, n]$$





Definitions: Assume that $y = f(x)$ is continuous.

- (a) The improper integral of f over the interval $[a, \infty)$ is given by

$$\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx ; [a, \infty) = \lim_{b \rightarrow \infty} [a, b]$$

- (b) The improper integral of f over the interval $(-\infty, b]$ is given by

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx ; (-\infty, b] = \lim_{a \rightarrow -\infty} [a, b]$$

- (c) The improper integral of f over the interval $(-\infty, \infty)$ is given by

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx, \text{ for some } c, -\infty < c < \infty. \\ &= \lim_{a \rightarrow -\infty} \int_a^c f(x) dx + \lim_{b \rightarrow \infty} \int_c^b f(x) dx. \end{aligned}$$

In the case where the limit exists, the improper integral is said to converge, and the limit is defined to be the value of the integral. In the case where the limit does not exist, the improper integral is said to diverge, and it is not assigned a value.

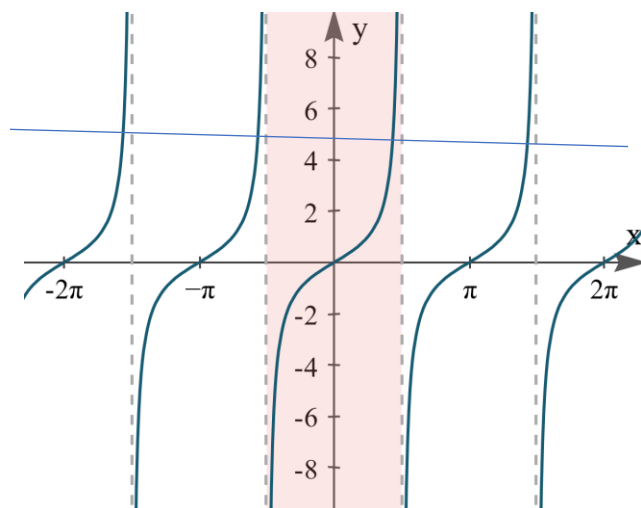
Theorem:

$$\int_1^{\infty} \frac{1}{x^p} dx = \begin{cases} \frac{1}{1-p} & \text{if } p < 1 \\ \text{diverges} & \text{if } p \geq 1. \end{cases}$$

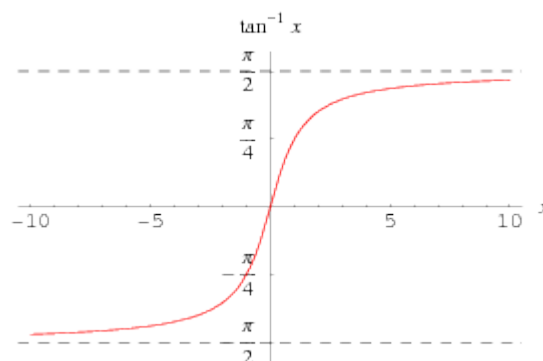
According to the theorem above, the integral $\int_1^{\infty} \frac{1}{x^2} dx$ diverges, and $\int_1^{\infty} \frac{1}{\sqrt{x}} dx$ converges.

Observation:

$$y = \tan x$$



$$y = \tan^{-1} x$$



$y = \tan x$ is **one-to-one** with the domain $D = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and range $R = (-\infty, \infty)$.

Hence, $y = \tan x$ has its inverse $y = \tan^{-1} x$ with the domain $D = (-\infty, \infty)$ and range $R = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$\lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2} \quad \text{and} \quad \lim_{x \rightarrow \infty} \tan^{-1} x = \frac{\pi}{2}$$

Exercise: 1

(a) Evaluate $\int_1^{\infty} \frac{1}{1+x^2} dx$.

Solution: The given integral $\int_1^{\infty} \frac{1}{1+x^2} dx$ is improper because of the infinite interval.

Then $\int_1^{\infty} \frac{1}{1+x^2} dx$

$$= \lim_{b \rightarrow \infty} \int_1^b \frac{1}{1+x^2} dx \quad ; \quad \left[\int_1^b \frac{1}{1+x^2} dx \text{ is a proper integral} \right]$$

$$= \lim_{b \rightarrow \infty} [\tan^{-1} x]_1^b$$

$$= \lim_{b \rightarrow \infty} [\tan^{-1} b - \tan^{-1} 1]$$

$$= \lim_{b \rightarrow \infty} \tan^{-1} b - \tan^{-1} 1$$

$$= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}, \text{ hence the integral is convergent.}$$

(b) Evaluate $\int_1^{\infty} e^{-2x} dx$.

Solution:

$$\begin{aligned}\int_1^{\infty} e^{-2x} dx &= \lim_{b \rightarrow \infty} \int_1^b e^{-2x} dx. \\&= \lim_{b \rightarrow \infty} \left[\frac{e^{-2x}}{-2} \right]_1^b = \lim_{b \rightarrow \infty} \left\{ -\frac{1}{2} [e^{-2b} - e^{-2}] \right\} \\&= \lim_{b \rightarrow \infty} \frac{-1}{2} e^{-2b} + \frac{1}{2} e^{-2} = \lim_{b \rightarrow \infty} \frac{-1}{2 e^{2b}} + \frac{1}{2} e^{-2} = 0 + \frac{1}{2e^2} \\&= \frac{1}{2e^2}, \text{ the integral converges.}\end{aligned}$$

Exercise: 2 Evaluate $\int_1^{\infty} e^{2x} dx$.

Solution:

$$\begin{aligned}\int_1^{\infty} e^{2x} dx &= \lim_{b \rightarrow \infty} \int_1^b e^{2x} dx \\&= \lim_{b \rightarrow \infty} \left[\frac{e^{2x}}{2} \right]_1^b = \lim_{b \rightarrow \infty} \frac{1}{2} [e^{2b} - e^2] \\&= \lim_{b \rightarrow \infty} \frac{1}{2} e^{2b} + \frac{1}{2} e^2 ; \quad \left(= \infty + \frac{e^2}{2} \right) \\&= \infty, \text{ the limit does not exist, hence the integral diverges.}\end{aligned}$$

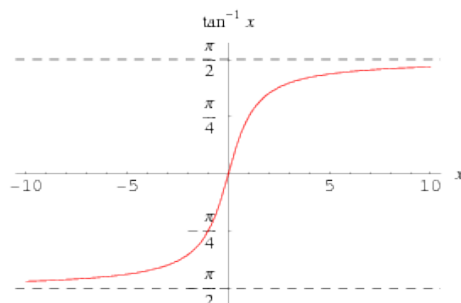
Exercise:3

Evaluate $\int_{-\infty}^0 \frac{1}{1+x^2} dx$.

Solution:

$$\begin{aligned}\int_{-\infty}^0 \frac{1}{1+x^2} dx &= \lim_{a \rightarrow -\infty} \int_a^0 \frac{1}{1+x^2} dx. \\&= \lim_{a \rightarrow -\infty} \left[\tan^{-1} x \right]_a^0 = \lim_{a \rightarrow -\infty} [\tan^{-1} 0 - \tan^{-1} a] \\&= \tan^{-1} 0 - \lim_{a \rightarrow -\infty} \tan^{-1} a = 0 - \left(-\frac{\pi}{2} \right) \\&= \frac{\pi}{2}, \text{ the integral converges.}\end{aligned}$$

Observation: $y = \tan^{-1} x$



$$\lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2} \quad \text{and} \quad \lim_{x \rightarrow \infty} \tan^{-1} x = \frac{\pi}{2}$$

Exercise:4

Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.

Hint: $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx$.

Exercise:5

Evaluate $\int_2^{\infty} \frac{1}{x\sqrt{\ln x}} dx$.

Solution: Given interval, $[2, \infty)$. Observe that the integral $\int_2^{\infty} \frac{1}{x\sqrt{\ln x}} dx$ is improper because of infinite interval.

$$\int_2^{\infty} \frac{1}{x\sqrt{\ln x}} dx$$

$$= \lim_{b \rightarrow \infty} \left[\int_2^b \frac{1}{x\sqrt{\ln x}} dx \right] \rightarrow \text{proper integral}$$

$$= \lim_{b \rightarrow \infty} \int_2^b \frac{1}{\sqrt{\ln x}} \cdot \frac{1}{x} dx.$$

set, $u = \ln x$, then $du = \frac{1}{x} dx$.

Also, if $x = 2$, then $u = \ln 2$ and if $x = b$, then $u = \ln b$

$$= \lim_{b \rightarrow \infty} \int_{\ln 2}^{\ln b} \frac{1}{\sqrt{u}} du$$

$$= \lim_{b \rightarrow \infty} \int_{\ln 2}^{\ln b} u^{-\frac{1}{2}} du$$

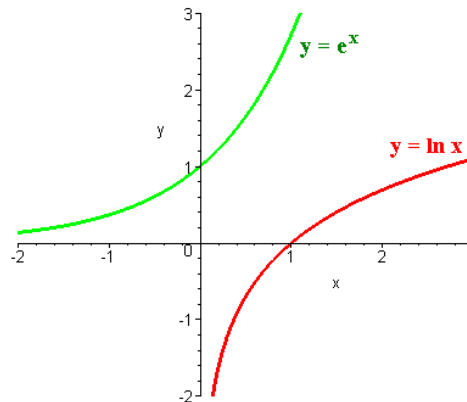
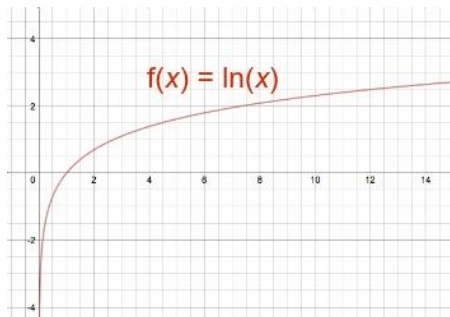
$$= \lim_{b \rightarrow \infty} \left[\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right]_{\ln 2}^{\ln b} = \lim_{b \rightarrow \infty} [2\sqrt{u}]_{\ln 2}^{\ln b} = \lim_{b \rightarrow \infty} [2\sqrt{\ln b} - 2\sqrt{\ln 2}]$$

$$= \lim_{b \rightarrow \infty} 2\sqrt{\ln b} - 2\sqrt{\ln 2}$$

$= \infty$, integral is divergent.

Note : if $x \rightarrow \infty$, then $y = \ln x \rightarrow \infty$. That is, $\lim_{x \rightarrow \infty} \ln x = \infty$

Note : if $x \rightarrow 0^+$, then $y = \ln x \rightarrow -\infty$. That is, $\lim_{x \rightarrow 0^+} \ln x = -\infty$



Exercise:6 [similar to ex 5] homework

Evaluate $\int_{-\infty}^{\infty} \frac{x}{(3+x^2)^2} dx$.

Hint: $\int_{-\infty}^{\infty} \frac{x}{(3+x^2)^2} dx = \int_{-\infty}^0 \frac{x}{(3+x^2)^2} dx + \int_0^{\infty} \frac{x}{(3+x^2)^2} dx$.

7.8: Improper Integrals

Proper Integrals: If $y = f(x)$ is continuous on some closed interval $[a, b]$, then $\int_a^b f(x) dx$ is called proper integral. [f is bounded on the interval].

Improper integrals:

- (4) Improper integrals with infinite intervals of integration:

$$\int_1^{\infty} \frac{1}{x^2} dx, \quad \int_{-\infty}^1 e^x dx, \quad \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$$

- (5) Improper integrals with infinite discontinuities in the intervals of integration:

$$\int_{-1}^1 \frac{1}{x^2} dx, \quad \int_1^2 \frac{1}{x-1} dx, \quad \int_0^{\pi} \tan x dx$$

- (6) Improper integrals with infinite discontinuities and infinite intervals of integration:

$$\int_{-\infty}^{\infty} \frac{1}{x^2} dx, \quad \int_{-\infty}^0 \frac{1}{\sqrt{x}} dx, \quad \int_{-\infty}^{\infty} \frac{1}{x^2-9} dx, \quad \int_1^{\infty} \sec x dx$$

Improper integrals with infinite discontinuities in the intervals of integration

➔ **Finitely many** discontinuities for vertical asymptotes over the given interval.

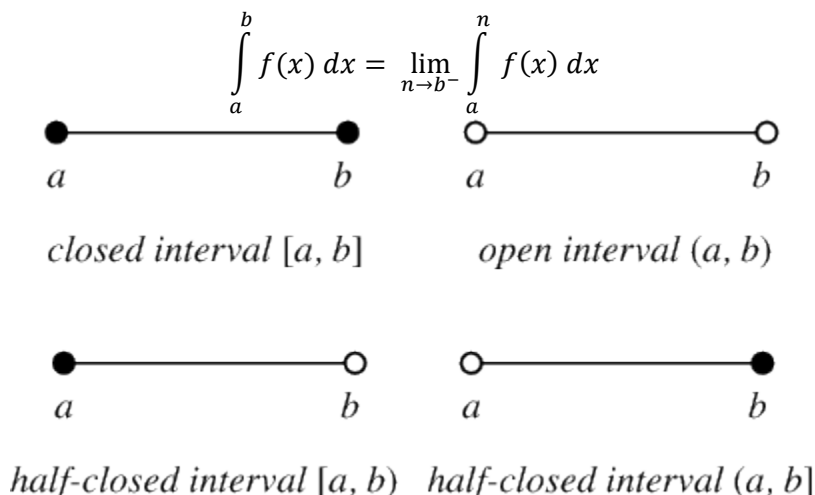
To solve an improper integral with infinite discontinuity, we need to use a closed interval with the help of limit concept where the function is continuous. For examples,

Definitions:

- (d) If the function $f(x)$ is **discontinuous only at $x = b$** on the interval $[a, b]$, then $f(x)$ is continuous on the interval $[a, b)$, and hence we get

$$[a, b) = \lim_{n \rightarrow b^-} [a, n] \quad ; \quad n < b, \text{ but close to } b, \quad \text{that is, } n \rightarrow b^-$$

Therefore,

$$\int_a^b f(x) dx = \lim_{n \rightarrow b^-} \int_a^n f(x) dx$$


$\underbrace{\quad}_{\text{closed interval } [a, b]} \quad \underbrace{\quad}_{\text{open interval } (a, b)}$
 $\underbrace{\quad}_{\text{half-closed interval } [a, b)} \quad \underbrace{\quad}_{\text{half-closed interval } (a, b]}$

- (e) If the function $f(x)$ is **discontinuous only at $x = a$** on the interval $[a, b]$, then $f(x)$ is continuous on the interval $(a, b]$, and hence we get

$$(a, b] = \lim_{n \rightarrow a^+} [n, b].$$

Therefore,

$$\int_a^b f(x) dx = \lim_{n \rightarrow a^+} \int_n^b f(x) dx.$$

- (f) If the function $f(x)$ is **discontinuous only at $x = a$ and $x = b$** on the interval $[a, b]$, then $f(x)$ is continuous on the interval (a, b) , and hence we get for some $a < c < b$.

$$(a, b) = (a, c] \cup [c, b) = \lim_{k \rightarrow a^+} [k, c] \cup \lim_{l \rightarrow b^-} [c, l].$$

Therefore,

$$\begin{aligned} \int_a^b f(x) dx &= \int_a^c f(x) dx + \int_c^b f(x) dx, \text{ for some } c, \quad a < c < b. \\ &= \lim_{k \rightarrow a^+} \int_k^c f(x) dx + \lim_{l \rightarrow b^-} \int_c^l f(x) dx. \end{aligned}$$

Note: We know that $\int_c^c f(x) dx = 0$.

(g) If the function $f(x)$ is **discontinuous only at $x = d$** for some $a < d < b$ on the interval $[a, b]$, then $f(x)$ is continuous on $[a, d) \cup (d, b]$, and hence we get

$$[a, d) \cup (d, b] = \lim_{k \rightarrow d^-} [a, k] \cup \lim_{l \rightarrow d^+} [l, b].$$

Therefore,

$$\begin{aligned} \int_a^b f(x) dx &= \int_a^d f(x) dx + \int_d^b f(x) dx \\ &= \lim_{k \rightarrow d^-} \int_a^k f(x) dx + \lim_{l \rightarrow d^+} \int_l^b f(x) dx. \end{aligned}$$

In the case where the limit exists, the improper integral is said to converge, and the limit is defined to be the value of the integral. In the case where the limit does not exist, the improper integral is said to diverge, and it is not assigned a value.

Exercise: 1

(a) Evaluate $\int_1^3 \frac{1}{(3-x)^2} dx$.

Solution: $f(x) = \frac{1}{(3-x)^2}$ is discontinuous only at $x = 3 \Rightarrow [1, 3) = \lim_{b \rightarrow 3^-} [1, b]$.

$$\begin{aligned} \int_1^3 \frac{1}{(3-x)^2} dx &= \lim_{b \rightarrow 3^-} \int_1^b \frac{1}{(3-x)^2} dx. \quad [\text{Set } u = 3 - x \Rightarrow -du = dx. \text{ Also, } 2 \leq u \leq 3 - b. \\ &= \lim_{b \rightarrow 3^-} \int_2^{3-b} \frac{1}{u^2} (-1) du = - \lim_{b \rightarrow 3^-} \int_2^{3-b} u^{-2} du = - \lim_{b \rightarrow 3^-} \left[\frac{u^{-1}}{-1} \right]_2^{3-b} \\ &= \lim_{b \rightarrow 3^-} \left[\frac{1}{u} \right]_2^{3-b} = \lim_{b \rightarrow 3^-} \left[\frac{1}{3-b} - \frac{1}{2} \right] = +\infty, \text{ the integral diverges.} \end{aligned}$$

(b) Evaluate

$$\int_{-3}^1 \frac{1}{(1-x)^{\frac{1}{2}}} dx.$$

Solution: Given $\int_{-3}^1 \frac{1}{(1-x)^{\frac{1}{2}}} dx$. $D = (-\infty, 1)$.

The function $f(x) = \frac{1}{(1-x)^{\frac{1}{2}}}$ is discontinuous at $x = 1$. So, the interval $I = [-3, 1)$

Now,

$$\int_{-3}^1 \frac{1}{(1-x)^{\frac{1}{2}}} dx$$

$$= \lim_{a \rightarrow 1^-} \int_{-3}^a \frac{1}{(1-x)^{\frac{1}{2}}} dx \leftarrow \text{proper integral}$$

[Set $u = 1 - x$. Then $(-1)du = dx$. Also, if $x = a$, then $u = 1 - a$ and if $x = -3$, then $u = 4$]

$$= \lim_{a \rightarrow 1^-} \int_4^{1-a} \frac{1}{u^{\frac{1}{2}}} (-1)du$$

$$= \lim_{a \rightarrow 1^-} \int_{1-a}^4 u^{-\frac{1}{2}} du \quad ; \quad - \int_b^a f(x)dx = \int_a^b f(x)dx$$

$$= \lim_{a \rightarrow 1^-} \left[\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right]_{1-a}^4$$

$$= \lim_{a \rightarrow 1^-} [2\sqrt{u}]_{1-a}^4$$

$$= \lim_{a \rightarrow 1^-} [2\sqrt{4} - 2\sqrt{1-a}]$$

$$= 2\sqrt{4} - \lim_{a \rightarrow 1^-} 2\sqrt{1-a}$$

$$= 4 - \lim_{a \rightarrow 1^-} 2\sqrt{1-a}$$

$$= 4 - 0$$

$$= 4, \text{ the integral converges.}$$

Note that $\lim_{a \rightarrow 1^+} 2\sqrt{1-a}$ is undefined.

(c) Evaluate

$$\int_1^2 \frac{1}{(2-x)^{\frac{2}{3}}} dx.$$

Solution: Given function $f(x) = \frac{1}{(2-x)^{\frac{2}{3}}}$ is discontinuous at $x = 2$. Hence, the given interval is $[1, 2)$.

Now,

$$\int_1^2 \frac{1}{(2-x)^{\frac{2}{3}}} dx$$

$$= \lim_{b \rightarrow 2^-} \int_1^b \frac{1}{(2-x)^{\frac{2}{3}}} dx \leftarrow \text{proper integral}$$

[Set $u = 2 - x$. Then $(-1)du = dx$. If $x = 1$, then $u = 1$ and if $x = b$, then $u = 2 - b$.]

$$= \lim_{b \rightarrow 2^-} \int_1^{2-b} \frac{1}{u^{\frac{2}{3}}} (-1) du$$

$$= \lim_{b \rightarrow 2^-} \int_{2-b}^1 u^{-\frac{2}{3}} du \quad ; \quad - \int_b^a f(x) dx = \int_a^b f(x) dx$$

$$= \lim_{b \rightarrow 2^-} \left[\frac{u^{\frac{1}{3}}}{\frac{1}{3}} \right]_{2-b}^1$$

$$= \lim_{b \rightarrow 2^-} [3 \sqrt[3]{u}]_{2-b}^1$$

$$= \lim_{b \rightarrow 2^-} [3 \sqrt[3]{1} - 3 \sqrt[3]{2-b}]$$

$$= 3 \sqrt[3]{1} - \lim_{b \rightarrow 2^-} 3 \sqrt[3]{2-b}$$

$$= 3 - 0$$

$$= 3, \text{ hence the integral is convergent.}$$

(d) Evaluate

$$\int_{-3}^1 \frac{x}{\sqrt{9-x^2}} dx.$$

Solution: Note that the given function $f(x) = \frac{x}{\sqrt{9-x^2}}$ is discontinuous at $x = -3$. So, we get the interval $I = (-3, 1]$. Hence, we have

$$\int_{-3}^1 \frac{x}{\sqrt{9-x^2}} dx = \lim_{a \rightarrow -3^+} \int_a^1 \frac{x}{\sqrt{9-x^2}} dx$$

$$= \lim_{a \rightarrow -3^+} \int_{9-a^2}^8 \frac{1}{\sqrt{u}} \left(-\frac{1}{2}\right) du \quad ; \quad [\text{Set } u = 9 - x^2. \text{ Then } \frac{du}{dx} = -2x, \quad \text{that is, } -\frac{1}{2} du = x dx.]$$

If $x = a$, then $u = 9 - a^2$, if $x = 1$, then $u = 8$]

$$= -\frac{1}{2} \lim_{a \rightarrow -3^+} \int_{9-a^2}^8 u^{-\frac{1}{2}} du$$

$$= -\frac{1}{2} \lim_{a \rightarrow -3^+} \left[\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right]_{9-a^2}^8$$

$$= -\lim_{a \rightarrow -3^+} [\sqrt{u}]_{9-a^2}^8$$

$$= -\lim_{a \rightarrow -3^+} [\sqrt{8} - \sqrt{9-a^2}]$$

$$= -\sqrt{8} + \lim_{a \rightarrow -3^+} \sqrt{9-a^2}$$

$$= -\sqrt{8} + 0$$

$$= -2\sqrt{2}$$

Therefore,

$$\int_{-3}^1 \frac{x}{\sqrt{9-x^2}} dx = -2\sqrt{2}.$$

Hence, the given integral is a convergent integral.

(e) Evaluate

$$\int_0^2 \frac{x}{\sqrt{4-x^2}} dx.$$

Solution: Note that the domain of the function $f(x) = \frac{x}{\sqrt{4-x^2}}$ is $[-2, 2]$, and $f(x)$ is discontinuous at $x = 2$. So, we have the interval $I = [0, 2)$. Hence, we have

$$\int_0^2 \frac{x}{\sqrt{4-x^2}} dx = \lim_{b \rightarrow 2^-} \int_0^b \frac{x}{\sqrt{4-x^2}} dx ;$$

[Set $u = 4 - x^2$. Then, $-\frac{1}{2} du = x dx$. Also, if $x = 0$, then $u = 4$ and if $x = b$, then $u = 4 - b^2$.]

$$\begin{aligned}
&= \lim_{b \rightarrow 2^-} \int_4^{4-b^2} \frac{1}{\sqrt{u}} \left(-\frac{1}{2}\right) du \\
&= \frac{1}{2} \lim_{b \rightarrow 2^-} \int_{4-b^2}^4 u^{-\frac{1}{2}} du \\
&= \frac{1}{2} \lim_{b \rightarrow 2^-} \left[\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right]_{4-b^2}^4 \\
&= \lim_{b \rightarrow 2^-} [\sqrt{u}]_{4-b^2}^4 \\
&= \lim_{b \rightarrow 2^-} [\sqrt{4} - \sqrt{4-b^2}] \\
&= \lim_{b \rightarrow 2^-} [2 - \sqrt{4-b^2}] \\
&= 2 - \lim_{b \rightarrow 2^-} \sqrt{4-b^2} \\
&= 2 - 0 \\
&= 2, \text{ Hence the given integral is a convergent integral.}
\end{aligned}$$

(f) Evaluate

$$\int_0^3 \frac{1}{x^2-1} dx.$$

Solution:

$$\begin{aligned}
\int_0^3 \frac{1}{x^2-1} dx &= \int_0^1 \frac{1}{x^2-1} dx + \int_1^3 \frac{1}{x^2-1} dx. \\
&= \lim_{b \rightarrow 1^-} \int_0^b \frac{1}{x^2-1} dx + \lim_{a \rightarrow 1^+} \int_a^3 \frac{1}{x^2-1} dx \\
&= \frac{1}{2} \lim_{b \rightarrow 1^-} \int_0^b \frac{(x+1) - (x-1)}{(x+1)(x-1)} dx + \frac{1}{2} \lim_{a \rightarrow 1^+} \int_a^3 \frac{(x+1) - (x-1)}{(x+1)(x-1)} dx \\
&= \frac{1}{2} \lim_{b \rightarrow 1^-} \int_0^b \left[\frac{1}{x-1} - \frac{1}{x+1} \right] dx + \frac{1}{2} \lim_{a \rightarrow 1^+} \int_a^3 \left[\frac{1}{x-1} - \frac{1}{x+1} \right] dx
\end{aligned}$$

$$\left[\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C, \text{ that is, } \int \frac{1}{x \pm 1} dx = \ln|x \pm 1| + C \right]$$

Homework:

(g) Evaluate

$$\int_0^{\infty} \frac{1}{x(x^2 - 1)} dx.$$

Exercise: 2

Evaluate

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sin \theta}{\sqrt{1 - 2 \cos \theta}} d\theta.$$

Solution: Given interval $I = \left[\frac{\pi}{3}, \frac{\pi}{2} \right]$, which is in the first quadrant. The cosine function is a decreasing function on the given interval $\left[\frac{\pi}{3}, \frac{\pi}{2} \right]$.

[Note: The value of $1 - 2 \cos \theta$ over the interval $\left[\frac{\pi}{3}, \frac{\pi}{2} \right]$ is from 0 to 1.]

Note that the function $f(\theta) = \frac{\sin \theta}{\sqrt{1 - 2 \cos \theta}}$ is discontinuous **only** at $\theta = \frac{\pi}{3}$, since $\cos \frac{\pi}{3} = \frac{1}{2}$.

So, we have $\left(\frac{\pi}{3}, \frac{\pi}{2} \right] = \lim_{a \rightarrow \left(\frac{\pi}{3} \right)^+} \left[a, \frac{\pi}{2} \right]$

$$\text{Now, } \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sin \theta}{\sqrt{1 - 2 \cos \theta}} d\theta$$

$$= \lim_{a \rightarrow \left(\frac{\pi}{3} \right)^+} \int_a^{\frac{\pi}{2}} \frac{\sin \theta}{\sqrt{1 - 2 \cos \theta}} d\theta \leftarrow \text{proper}$$

[Set $u = 1 - 2 \cos \theta \Rightarrow \frac{du}{d\theta} = 2 \sin \theta \Rightarrow \frac{1}{2} du = \sin \theta d\theta$. Also, $1 - 2 \cos a \leq u \leq 1$]

$$= \lim_{a \rightarrow (\frac{\pi}{3})^+} \int_{1-2\cos a}^1 \frac{1}{\sqrt{u}} \cdot \frac{1}{2} du$$

$$= \lim_{a \rightarrow (\frac{\pi}{3})^+} \frac{1}{2} \int_{1-2\cos a}^1 u^{-\frac{1}{2}} du$$

$$= \lim_{a \rightarrow (\frac{\pi}{3})^+} \left. \frac{1}{2} \cdot \frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right|_{1-2\cos a}^1$$

$$= \lim_{a \rightarrow (\frac{\pi}{3})^+} \sqrt{u} \Big|_{1-2\cos a}^1$$

$$= \lim_{a \rightarrow (\frac{\pi}{3})^+} [1 - \sqrt{1 - 2\cos a}] ;$$

[Since $a > \frac{\pi}{3}$, and the cosine is a **decreasing** function over the interval $[\frac{\pi}{3}, \frac{\pi}{2}]$, so we get **$\cos a < \frac{1}{2}$** .]

$= 1 - 0 = 1$, the integral converges.

Definitions:

Decreasing Function

1) A function $y = f(x)$ is said to be decreasing if and only if $x_1 \geq x_2$ implies that $f(x_1) \leq f(x_2)$.

Increasing Function

2) A function $y = f(x)$ is said to be increasing if and only if $x_1 \geq x_2$ implies that $f \geq f(x_2)$.

Exercise:3 Evaluate $\int_0^3 \frac{1}{x-2} dx$. **homework**

The function is discontinuous at $x = 2$.

Hint: $\int_0^3 \frac{1}{x-2} dx = \int_0^2 \frac{1}{x-2} dx + \int_2^3 \frac{1}{x-2} dx$. Understanding of Exercises 1(a) and 2

Logarithmic Derivative:

$$\frac{d}{dx} [\ln(f(x))] = \frac{f'(x)}{f(x)} \leftarrow \text{Logarithmic Derivative} = \frac{\text{Derivative of the denominator}}{\text{Denominator}}$$

We know that

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

For example,

$$\int \frac{1}{x-2} dx = \ln |x-2| + C$$

Exercise:4 (a) Evaluate $\int_0^{\infty} \frac{1}{x^2} dx$.

Solution: The function $f(x) = \frac{1}{x^2}$ is discontinuous only at $x = 0$ over the infinite interval $(0, \infty)$.

$$\int_0^{\infty} \frac{1}{x^2} dx = \int_0^1 \frac{1}{x^2} dx + \int_1^{\infty} \frac{1}{x^2} dx$$

$$= \lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{x^2} dx + \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^2} dx$$

$$= \lim_{a \rightarrow 0^+} \int_a^1 x^{-2} dx + \lim_{b \rightarrow \infty} \int_1^b x^{-2} dx$$

$$= \lim_{a \rightarrow 0^+} [-x^{-1}]_a^1 + \lim_{b \rightarrow \infty} [-x^{-1}]_1^b$$

$$= \lim_{a \rightarrow 0^+} \left[-\frac{1}{x}\right]_a^1 + \lim_{b \rightarrow \infty} \left[-\frac{1}{x}\right]_1^b$$

$$= \lim_{a \rightarrow 0^+} \left[-\frac{1}{1} + \frac{1}{a}\right] + \lim_{b \rightarrow \infty} \left[-\frac{1}{b} + \frac{1}{1}\right]$$

$$= \lim_{a \rightarrow 0^+} \frac{1}{a} - \lim_{b \rightarrow \infty} \frac{1}{b}$$

$= \infty$, Hence the integral is divergent.

(b) Evaluate

$$\int_{-\infty}^0 \frac{x}{\sqrt{1-x^2}} dx.$$

Solution: Problem at $x = 1$ and at $-\infty$

(c) Evaluate

$$\int_0^{\infty} \frac{1}{x^3} dx.$$

Solution:

$$\begin{aligned}\int_0^{\infty} \frac{1}{x^3} dx &= \int_0^1 \frac{1}{x^3} dx + \int_1^{\infty} \frac{1}{x^3} dx \\&= \lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{x^3} dx + \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^3} dx \\&= \lim_{a \rightarrow 0^+} \int_a^1 x^{-3} dx + \lim_{b \rightarrow \infty} \int_1^b x^{-3} dx \\&= \lim_{a \rightarrow 0^+} \left[-\frac{1}{2} x^{-2} \right]_a^1 + \lim_{b \rightarrow \infty} \left[-\frac{1}{2} x^{-2} \right]_1^b \\&= \lim_{a \rightarrow 0^+} \left[-\frac{1}{2x^2} \right]_a^1 + \lim_{b \rightarrow \infty} \left[-\frac{1}{2x^2} \right]_1^b \\&= \lim_{a \rightarrow 0^+} \left[-\frac{1}{2} + \frac{1}{2a^2} \right] + \lim_{b \rightarrow \infty} \left[-\frac{1}{2b^2} + \frac{1}{2} \right] \\&= -\frac{1}{2} + \lim_{a \rightarrow 0^+} \frac{1}{2a^2} - \lim_{b \rightarrow \infty} \frac{1}{2b^2} + \frac{1}{2} \\&= \lim_{a \rightarrow 0^+} \frac{1}{2a^2} - \lim_{b \rightarrow \infty} \frac{1}{2b^2} \\&= \infty, \text{ So the integral diverges.}\end{aligned}$$

Homework

(d) Evaluate

$$\int_0^{\infty} \frac{x}{x^2 - 1} dx.$$

Solution: Problem at $x = 1$ and at ∞ .

$$\begin{aligned} \int_0^{\infty} \frac{x}{x^2 - 1} dx &= \int_0^1 \frac{x}{x^2 - 1} dx + \int_1^{\infty} \frac{x}{x^2 - 1} dx \\ &= \int_0^1 \frac{x}{x^2 - 1} dx + \int_1^2 \frac{x}{x^2 - 1} dx + \int_2^{\infty} \frac{x}{x^2 - 1} dx \\ &= \lim_{b \rightarrow 1^-} \int_0^b \frac{x}{x^2 - 1} dx + \lim_{a \rightarrow 1^+} \int_a^2 \frac{x}{x^2 - 1} dx + \lim_{k \rightarrow \infty} \int_2^k \frac{x}{x^2 - 1} dx \end{aligned}$$